UNIVERSITY CAMPUS NETWORK USING CISCO PACKET TRACER

Network Design, Configuration, Simulation and Testing — Final Report

Field	Details
Simulation Platform	Cisco Packet Tracer 9.0.0
Project File	University_Campus_Network_Premier_University.pkt

Contents

(Right-click the table and choose Update Field in Microsoft Word to refresh page numbers.)

1. Abstract

This report documents the design, configuration and testing of a simulated smart university campus network for Premier University, Chittagong, built directly in Cisco Packet Tracer 9.0. The network contains 31 devices: a simulated ISP and Internet server, a Cisco 2911 edge router, two Catalyst 3650 multilayer distribution switches, five Catalyst 2960 access switches, four data-centre servers, seven PCs, a WPA2 access point with two wireless clients, and four IoT devices. Nine VLANs are addressed from 10.10.0.0/16 with VLSM, routed with OSPFv2 (IPv4) and OSPFv3 (IPv6), served by a central DHCP/DNS server through DHCP relay, protected by three named extended ACLs, an SSH-only management policy and NAT/PAT, and extended with an IoT registration server that switches classroom lighting by occupancy. Of 31 planned tests, 23 passed, 2 failed and 6 were not run; every result is reported as observed, with screenshots captured from the running simulation.

2. Introduction

University campuses combine user groups with very different needs — students, registrar and administrative staff, faculty, researchers, facilities and IT — on one shared infrastructure. The project brief for CSE 3568 asks for a comprehensive network that demonstrates multi-zone routing, VLAN segmentation, VLSM and IPv6 addressing, wireless access, ACL and NAT security, and an IoT automation function. The smart-city scenario in the course template was adapted to a smart campus, as described in the approved-for-review proposal. All campus segments are simulated; they are not a description of the university's real infrastructure.

3. Problem Statement

Design and build, in Cisco Packet Tracer, a campus network in which: (i) every authorised segment can reach the services it needs; (ii) students cannot reach administrative, management or IoT segments; (iii) clients are addressed automatically from a central server; (iv) the campus reaches a simulated Internet through a single translated public address while only the web portal is published inward; (v) at least one segment runs IPv6 end to end; (vi) students can connect over secured Wi-Fi; and (vii) a smart-campus automation rule reacts to a sensor condition.

4. Objectives

- Build a routed two-block campus with OSPF adjacencies reaching FULL state between all Layer-3 devices.
- Implement a non-overlapping VLSM plan for nine VLANs and three /30 links, plus dual-stack IPv6 on the library and server VLANs.
- Enforce an inter-zone security policy with named ACLs, an SSH-only management ACL and NAT/PAT, verified by permit and deny tests.
- Provide WPA2-AES student Wi-Fi and centralised DHCP/DNS through DHCP relay.
- Demonstrate occupancy-based lighting and temperature-based ventilation rules on a central IoT server.

5. Scope and Assumptions

- All segments, addresses and services are simulated; public addresses use RFC 5737 (203.0.113.0/29, 198.51.100.0/24) and IPv6 uses the RFC 3849 documentation prefix 2001:DB8::/32.
- The two building blocks are joined by routed links in one OSPF area; there are no WAN serial links between campuses.
- IPv6 is internal only (VLAN 50, VLAN 60 and link L3); no IPv6 Internet service is simulated.
- Lab-only credentials (netadmin / PULab2026, Wi-Fi PUStudent2026, IoT iotadmin / PUIoT2026) are used for demonstration and must be changed before any reuse.
- Packet Tracer feature support defines the limits of the design; differences from the proposal are listed in Section 22.

6. Requirements

ID	Requirement	Status
FR1	Inter-zone routing with OSPF and a redundant inter-block path	Implemented; adjacencies verified (failover not tested)
FR2	VLAN segmentation, 802.1Q trunks with native VLAN 999, SVI inter-VLAN routing	Implemented and verified by DHCP/ping tests
FR3	Central DHCP (relay) and DNS	Implemented and verified
FR4	WPA2-AES student wireless in VLAN 40	Implemented; association and DHCP verified
FR5	ACLs, SSH-only management, NAT/PAT and static NAT	Implemented and verified (some negative tests not run)
FR6	Dual-stack IPv6 with OSPFv3; IoT automation	Implemented; IPv6 ping and motion rule verified; temperature rule not triggered

7. Network Architecture

The design is a collapsed core/distribution hierarchy in two blocks. R-EDGE (Cisco 2911) is the only Internet boundary and performs NAT/PAT and inbound filtering. Two Catalyst 3650-24PS multilayer switches act as the distribution and inter-VLAN routing layer for the Academic & Library block (DSW-ACAD) and the Administration & Data Center block (DSW-ADMIN). The three Layer-3 devices form a triangle of routed /30 links (L1, L2, L3) in OSPF Area 0, so traffic between blocks normally uses L3 and can fall back through R-EDGE. VLANs are local to each block, which keeps every spanning-tree domain small. Access is provided by five Catalyst 2960-24TT switches.

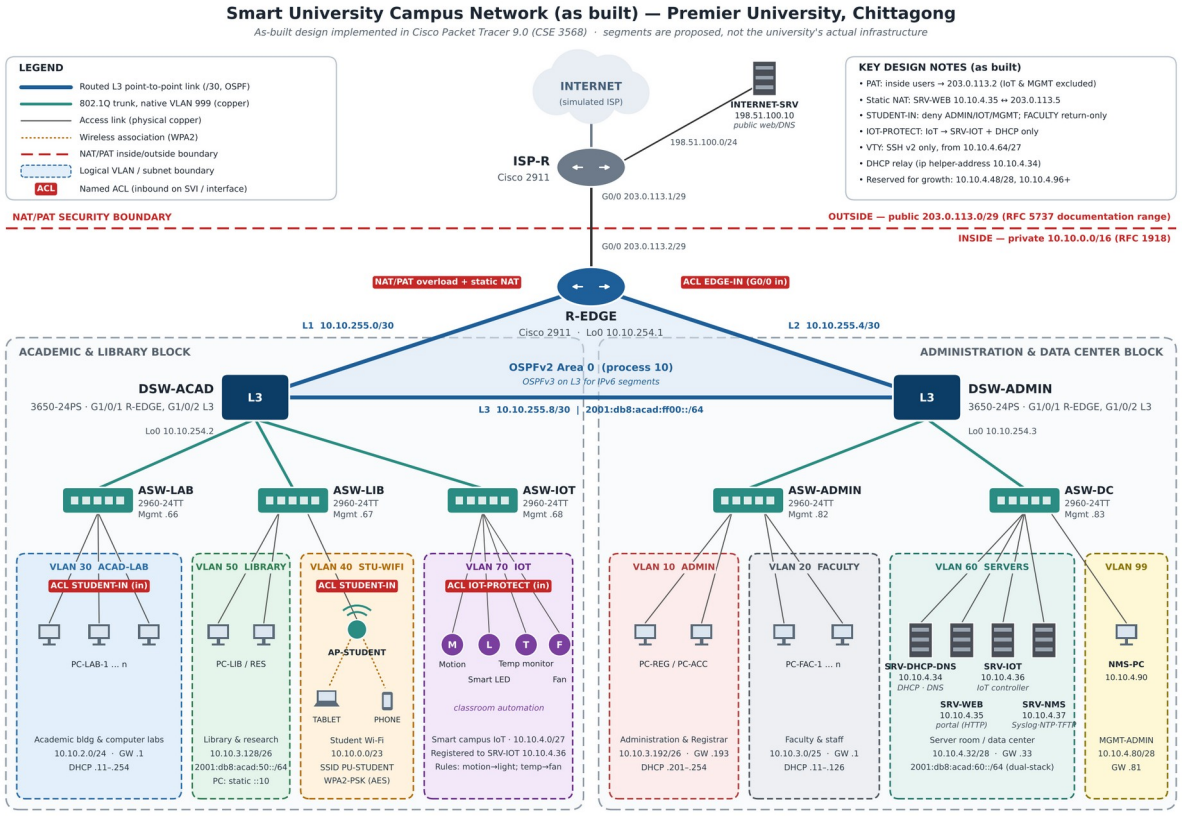


Figure 1. As-built logical design (diagram drawn from the implemented configuration).

8. Actual Packet Tracer Topology

Figure 2 is a screenshot of the logical workspace of the saved project file. Device labels match the configured hostnames; port labels show the cabling used.

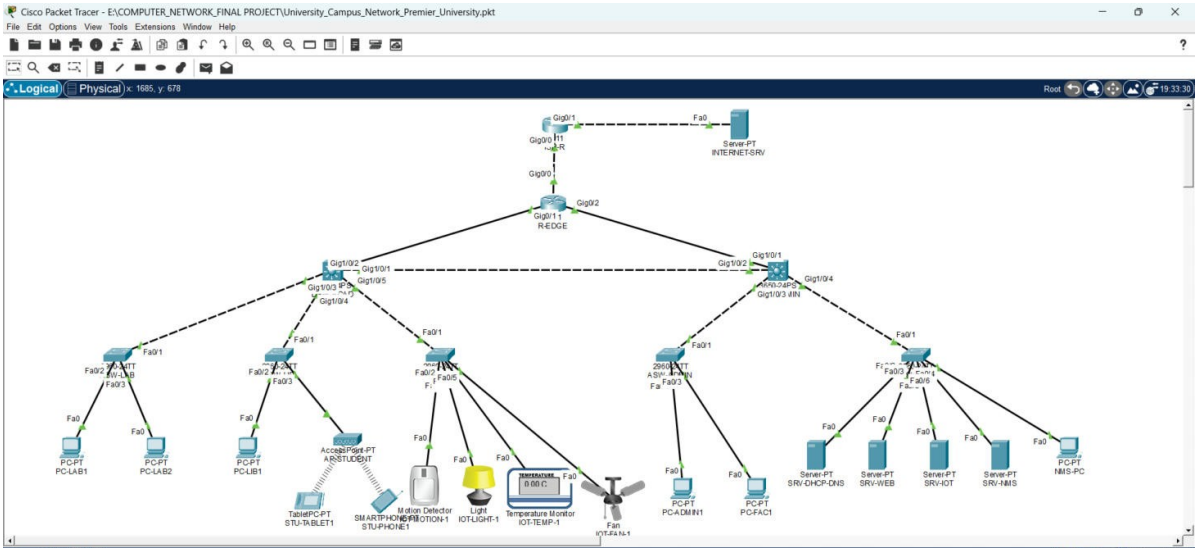


Figure 2. Logical topology in Cisco Packet Tracer 9.0 (screenshot of the saved project).

9. Device Inventory

Device	Model	Role	Key interfaces / address
ISP-R	Cisco 2911	Simulated ISP	G0/0 203.0.113.1/29, G0/1 198.51.100.1/24
INTERNET-SRV	Server-PT	Simulated Internet web server	198.51.100.10/24
R-EDGE	Cisco 2911	Edge router, NAT/PAT, EDGE-IN ACL, OSPF ASBR	G0/0 203.0.113.2/29, G0/1 10.10.255.1/30, G0/2 10.10.255.5/30, Lo0 10.10.254.1
DSW-ACAD	Catalyst 3650-24PS	Distribution, SVIs 30/40/50/70/98	G1/0/1 10.10.255.2, G1/0/2 10.10.255.9, trunks G1/0/3-5
DSW-ADMIN	Catalyst 3650-24PS	Distribution, SVIs 10/20/60/99	G1/0/1 10.10.255.6, G1/0/2 10.10.255.10, trunks G1/0/3-4
ASW-LAB / ASW-LIB / ASW-IOT	Catalyst 2960-24TT	Access (academic block)	Fa0/1 trunk; mgmt 10.10.4.66/.67/.68
ASW-ADMIN / ASW-DC	Catalyst 2960-24TT	Access (admin block)	Fa0/1 trunk; mgmt 10.10.4.82/.83
SRV-DHCP-DNS / SRV-WEB / SRV-IOT / SRV-NMS	Server-PT	DHCP+DNS / portal / IoT controller / syslog	10.10.4.34 / .35 / .36 / .37
AP-STUDENT	AccessPoint-PT	WPA2-AES SSID PU-STUDENT	ASW-LIB Fa0/3, VLAN 40
PC-LAB1, PC-LAB2, PC-LIB1, PC-ADMIN1, PC-FAC1, NMS-PC	PC-PT	Representative users	DHCP (NMS-PC static 10.10.4.90)
STU-TABLET1, STU-PHONE1	TabletPC-PT, Smartphone-PT	Wireless clients	DHCP 10.10.0.11, 10.10.0.13
IOT-MOTION-1, IOT-LIGHT-1, IOT-TEMP-1, IOT-FAN-1	Home IoT devices	Sensors and actuators	Wired adapter PT-IOT-NM-1CFE, VLAN 70

10. VLAN Table

VLAN	Name	Purpose	Switches	Gateway (SVI)
10	ADMIN	Administration & Registrar	DSW-ADMIN, ASW-ADMIN	10.10.3.193
20	FACULTY	Faculty & staff	DSW-ADMIN, ASW-ADMIN	10.10.3.1
30	ACAD-LAB	Computer labs	DSW-ACAD, ASW-LAB	10.10.2.1
40	STU-WIFI	Student wireless	DSW-ACAD, ASW-LIB	10.10.0.1
50	LIBRARY	Library & research (dual-stack)	DSW-ACAD, ASW-LIB	10.10.3.129
60	SERVERS	Data centre (dual-stack)	DSW-ADMIN, ASW-DC	10.10.4.33
70	IOT	Smart campus devices	DSW-ACAD, ASW-IOT	10.10.4.1

VLAN	Name	Purpose	Switches	Gateway (SVI)
98	MGMT-ACAD	Switch management (academic)	DSW-ACAD + academic ASWs	10.10.4.65
99	MGMT-ADMIN	Switch management + NMS-PC	DSW-ADMIN + admin ASWs	10.10.4.81
998	PARKING	Unused, shut-down ports	All switches	—
999	NATIVE	Unused native VLAN on trunks	All trunks	—

11. IPv4 and VLSM Addressing Table

Subnets were allocated largest first from 10.10.0.0/16. Every subnet was checked by script for overlap and DHCP-scope containment before configuration. VLANs 98 and 99 occupy one aligned /27 (10.10.4.64/27), so one ACL wildcard (0.0.0.31) matches both management VLANs. 10.10.4.96 – 10.10.253.255 remains free.

VLAN/ Link	Name	Network	Prefix	Subnet mask	First usable	Last usable	Broadcast	DHCP range
10	ADMIN (Administration & Registrar)	10.10.3.192	/26	255.255.255.192	10.10.3.193	10.10.3.254	10.10.3.255	10.10.3.201-10.10.3.254
20	FACULTY (Faculty & Staff)	10.10.3.0	/25	255.255.255.128	10.10.3.1	10.10.3.126	10.10.3.127	10.10.3.11-10.10.3.126
30	ACAD-LAB (Academic Bldg & Computer Labs)	10.10.2.0	/24	255.255.255.0	10.10.2.1	10.10.2.254	10.10.2.255	10.10.2.11-10.10.2.254
40	STU-WIFI (Student Wi-Fi)	10.10.0.0	/23	255.255.254.0	10.10.0.1	10.10.1.254	10.10.1.255	10.10.0.11-10.10.1.254
50	LIBRARY (Library & Research)	10.10.3.128	/26	255.255.255.192	10.10.3.129	10.10.3.190	10.10.3.191	10.10.3.136-10.10.3.190
60	SERVERS (Server Room / Data Center)	10.10.4.32	/28	255.255.255.240	10.10.4.33	10.10.4.46	10.10.4.47	static
70	IOT (Smart Campus IoT)	10.10.4.0	/27	255.255.255.224	10.10.4.1	10.10.4.30	10.10.4.31	10.10.4.11-10.10.4.30
98	MGMT-ACAD (Network Mgmt (Academic block))	10.10.4.64	/28	255.255.255.240	10.10.4.65	10.10.4.78	10.10.4.79	static
99	MGMT-ADMIN (Network Mgmt & Security)	10.10.4.80	/28	255.255.255.240	10.10.4.81	10.10.4.94	10.10.4.95	static
L1	R-EDGE G0/1 ↔ DSW-ACAD G1/0/1	10.10.255.0	/30	255.255.255.252	10.10.255.1	10.10.255.2	10.10.255.3	-
L2	R-EDGE G0/2 ↔ DSW-ADMIN G1/0/1	10.10.255.4	/30	255.255.255.252	10.10.255.5	10.10.255.6	10.10.255.7	-
L3	DSW-ACAD G1/0/2 ↔ DSW-ADMIN G1/0/2	10.10.255.8	/30	255.255.255.252	10.10.255.9	10.10.255.10	10.10.255.11	-
Lo0	OSPF router-IDs	10.10.254.1-3	/32	255.255.255.255				-
WAN	R-EDGE G0/0 - ISP-R G0/0	203.0.113.0	/29	255.255.255.248	203.0.113.1	203.0.113.6	203.0.113.7	-
ISP LAN	Simulated Internet	198.51.100.0	/24	255.255.255.0	198.51.100.1	198.51.100.254	198.51.100.255	-
Reserved	Future DMZ / server growth	10.10.4.48	/28	255.255.255.240				

12. IPv6 Configuration

Segment	Prefix	Router address	Host
VLAN 50 Library	2001:DB8:ACAD:50::/64	DSW-ACAD SVI ::1, link-local FE80::1	PC-LIB1 2001:DB8:ACAD:50::10 (static), gateway FE80::1
VLAN 60 Servers	2001:DB8:ACAD:60::/64	DSW-ADMIN SVI ::1, link-local FE80::1	SRV-WEB 2001:DB8:ACAD:60::35, gateway FE80::1
Link L3	2001:DB8:ACAD:FF00::/64	DSW-ACAD ::1, DSW-ADMIN ::2	—

ipv6 unicast-routing and OSPFv3 process 10 (Area 0) run on both 3650 switches. The OSPFv3 adjacency reached FULL and DSW-ACAD learned 2001:DB8:ACAD:60::/64 (Figure 9a). PC-LIB1 did not obtain a global address by SLAAC, so a static address was configured; the IPv6 ping to the web server then succeeded (Figure 15).

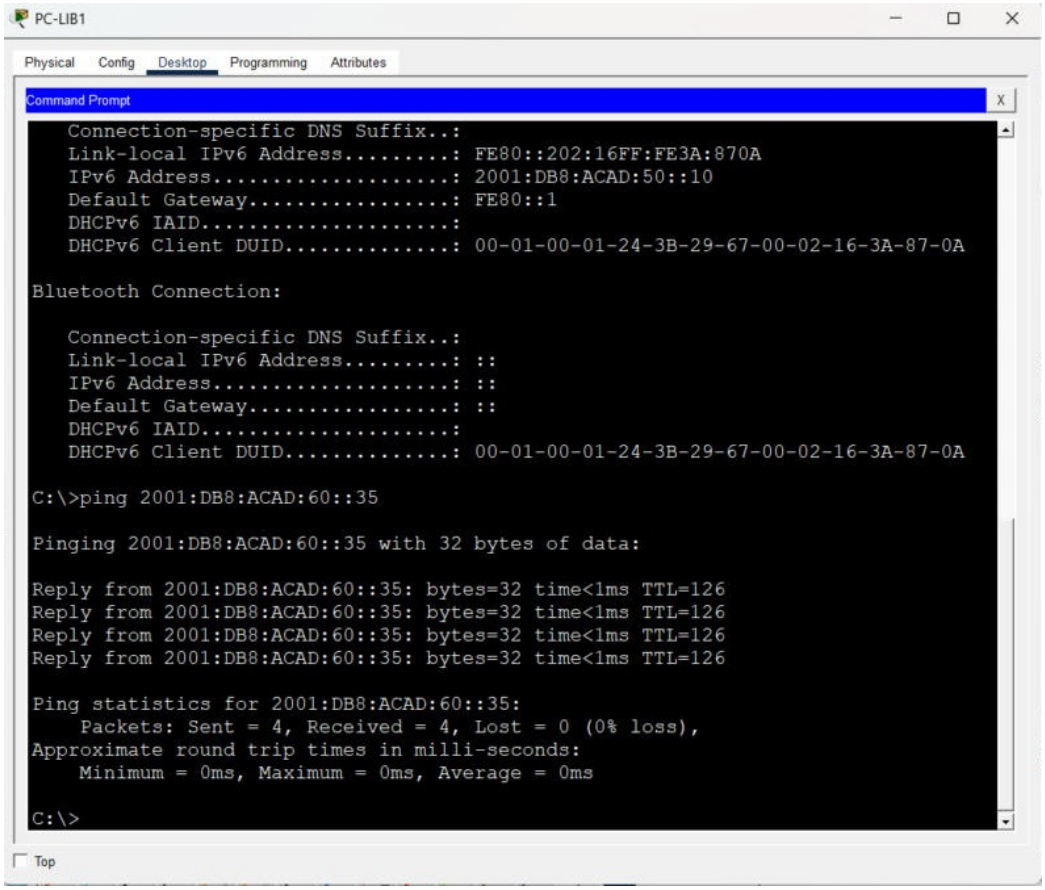


Figure 15. PC-LIB1: static IPv6 address and 0% loss ping to SRV-WEB 2001:DB8:ACAD:60::35.

13. Routing Design

OSPFv2 process 10 runs in a single Area 0 with loopback router-IDs 10.10.254.1–3. On the 3650s, passive-interface default is used and only the two routed uplinks are made active, so SVI subnets are advertised without forming adjacencies toward users. R-EDGE holds a static default route to the ISP and advertises it with default-information originate. OSPF was chosen over EIGRP (vendor-specific) and RIP (hop-count metric, slow convergence). The ISP router intentionally has no route to 10.10.0.0/16, so Internet replies can return only through NAT.

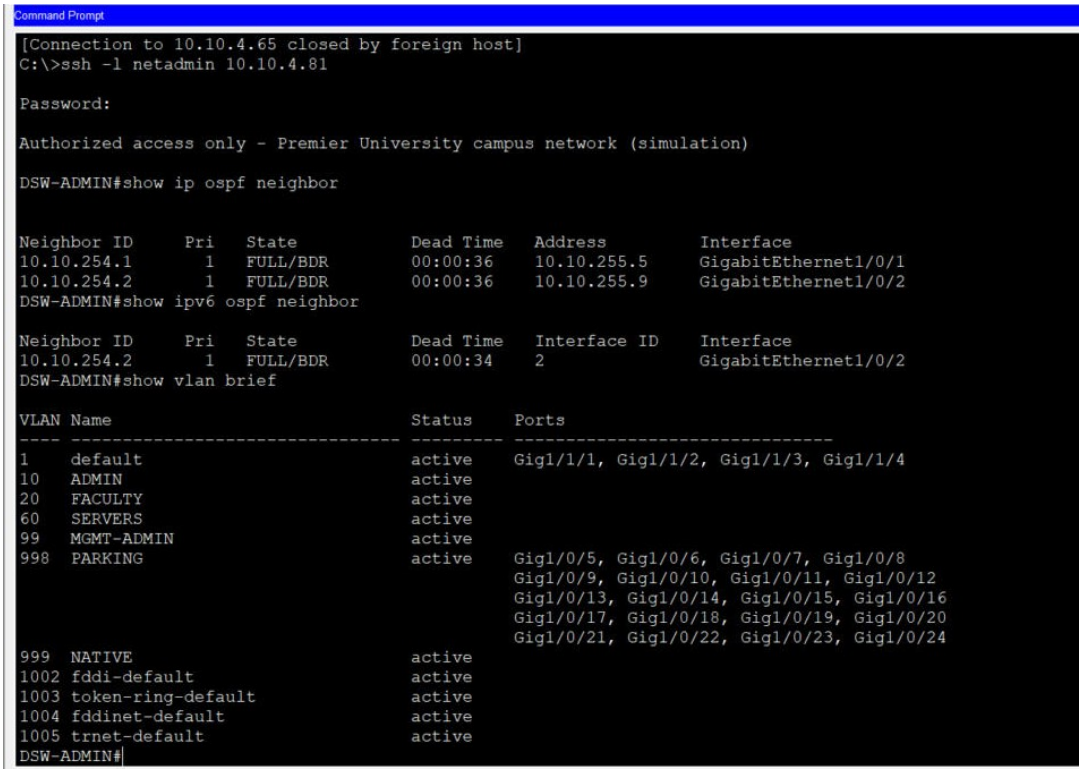


Figure 3. DSW-ADMIN: OSPFv2 neighbours (R-EDGE, DSW-ACAD) and OSPFv3 neighbour FULL; block VLANs.

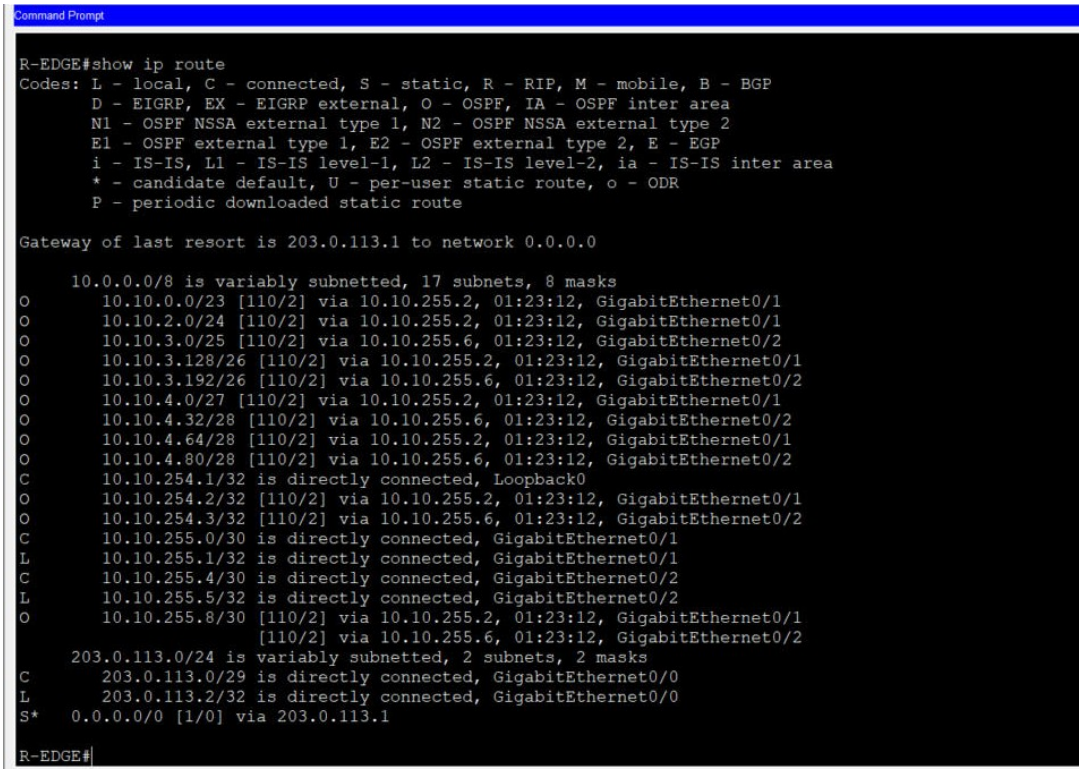


Figure 4. R-EDGE routing table: every campus subnet learned by OSPF, and the static default route.

14. DHCP and DNS

SRV-DHCP-DNS (10.10.4.34) holds one pool per client VLAN (10, 20, 30, 40, 50, 70). Each SVI forwards DHCP broadcasts with ip helper-address 10.10.4.34, and the server selects the pool that matches the relay address. DNS A records point portal.pu.internal, iot.pu.internal and nms.pu.internal at the servers, and an AAAA record (portal6.pu.internal) holds the portal's IPv6 address. Observed leases: PC-LAB1 10.10.2.11, PC-

LAB2 10.10.2.12, PC-ADMIN1 10.10.3.201, PC-FAC1 10.10.3.11, PC-LIB1 10.10.3.136, tablet 10.10.0.12 (10.10.0.11 after re-lease), phone 10.10.0.13, motion detector 10.10.4.11.

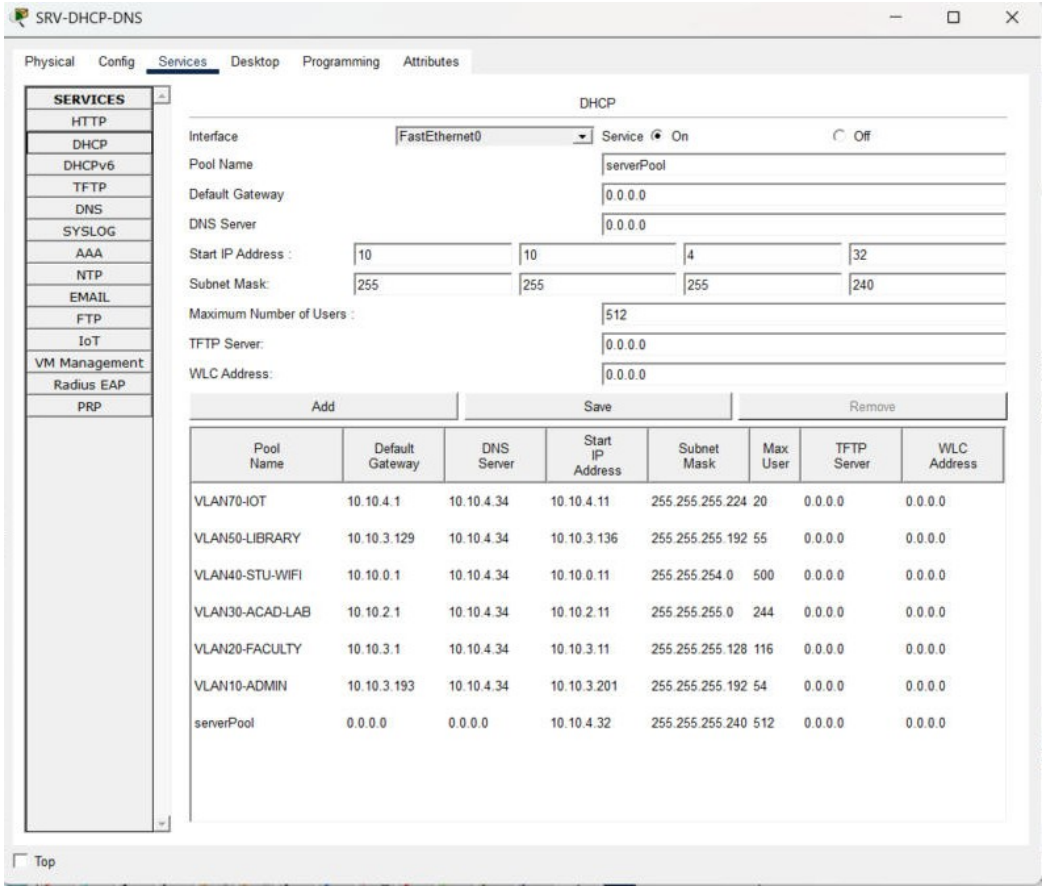


Figure 5a. DHCP pools on SRV-DHCP-DNS.

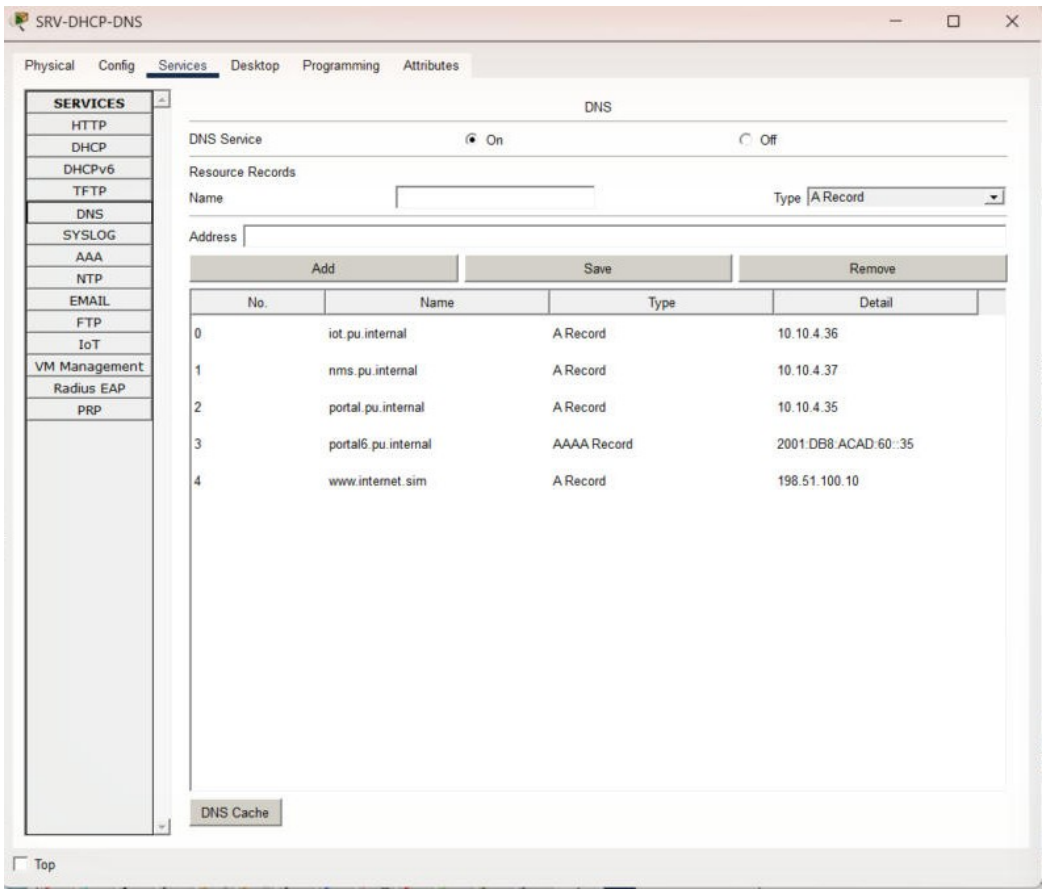


Figure 5b. DNS resource records.

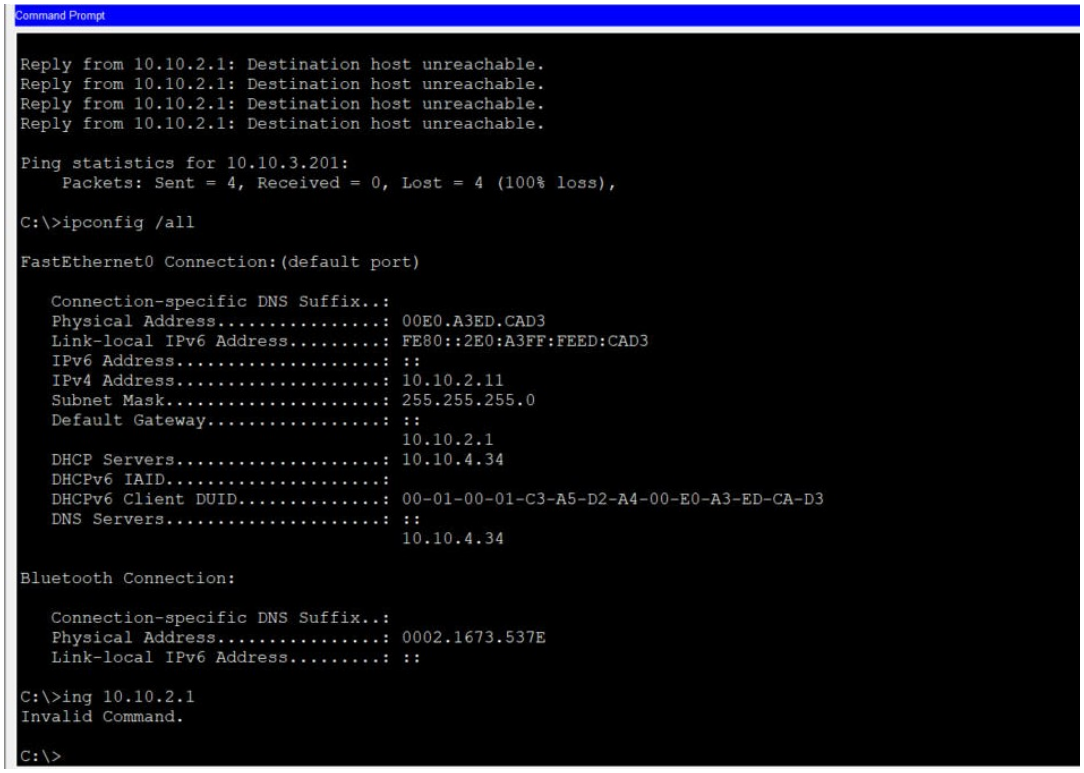


Figure 6. PC-LAB1 lease from the central server (10.10.4.34) through DHCP relay.

15. Server Configuration

SRV-WEB serves a campus portal page over HTTP on 10.10.4.35 and 2001:DB8:ACAD:60::35, and is published to the outside as 203.0.113.5 by static NAT. SRV-IOT runs the IoT registration service. SRV-NMS has its syslog service enabled; no network device has yet been configured to send logs to it (logging host is a recommended next step). INTERNET-SRV (198.51.100.10) represents a public website behind ISP-R.

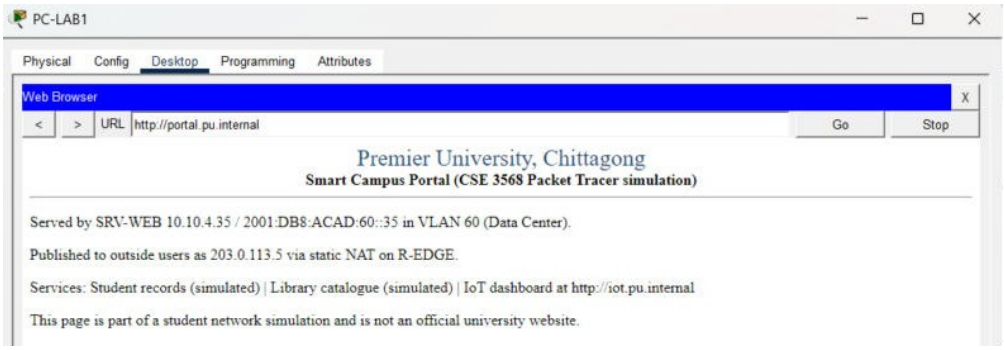


Figure 7. Campus portal opened by name from PC-LAB1 (DNS + inter-VLAN routing + HTTP).

16. ACL and NAT/PAT

Policy	Applied	Rules (summary)
STUDENT-IN (extended)	DSW-ACAD SVI 30 and SVI 40, inbound	Permit DHCP; permit established TCP and echo-replies to faculty; deny faculty, admin (10.10.3.192/26), IoT (10.10.4.0/27), management (10.10.4.64/27); permit everything else
IOT-PROTECT (extended)	DSW-ACAD SVI 70, inbound	Permit DHCP; permit IoT → SRV-IOT 10.10.4.36; permit echo-replies to management; deny all else
EDGE-IN (extended)	R-EDGE G0/0, inbound	Permit HTTP to 203.0.113.5; permit established TCP, DNS replies, echo-reply, unreachable; deny all else
MGMT-ONLY (standard)	access-class on VTY lines of all 8 network devices	Permit 10.10.4.64/27 only; VTY accepts SSH v2 only, local login

Policy	Applied	Rules (summary)
NAT-USERS + PAT	R-EDGE inside G0/1-2, outside G0/0	Deny IoT and management subnets; permit 10.10.0.0/16; overload on G0/0 (203.0.113.2)
Static NAT	R-EDGE	10.10.4.35 ↔ 203.0.113.5

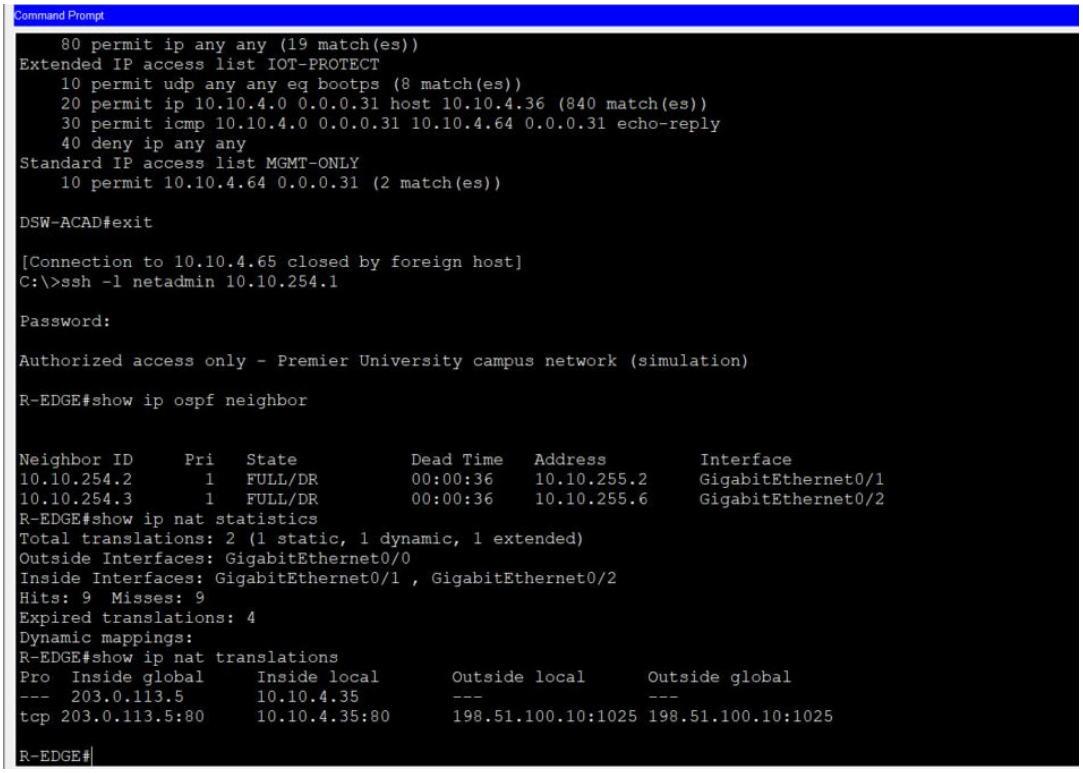


Figure 8a. R-EDGE (SSH from NMS-PC): OSPF neighbours, NAT statistics and translations.

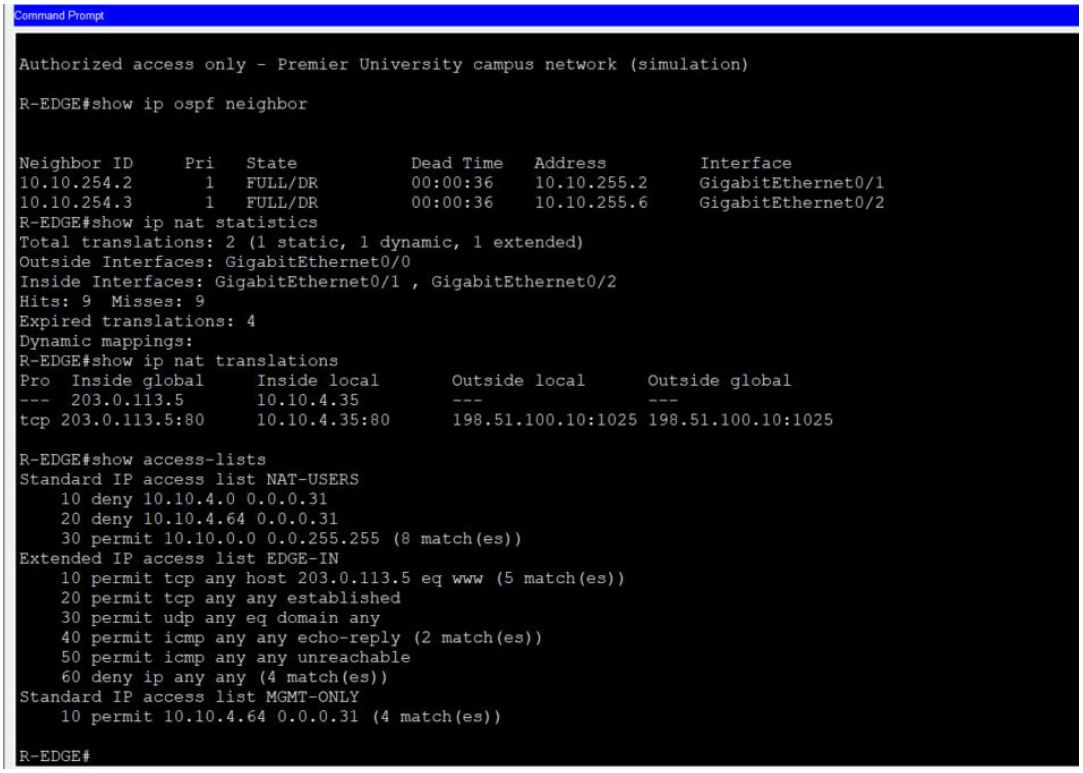


Figure 8b. R-EDGE: NAT-USERS, EDGE-IN and MGMT-ONLY hit counters.

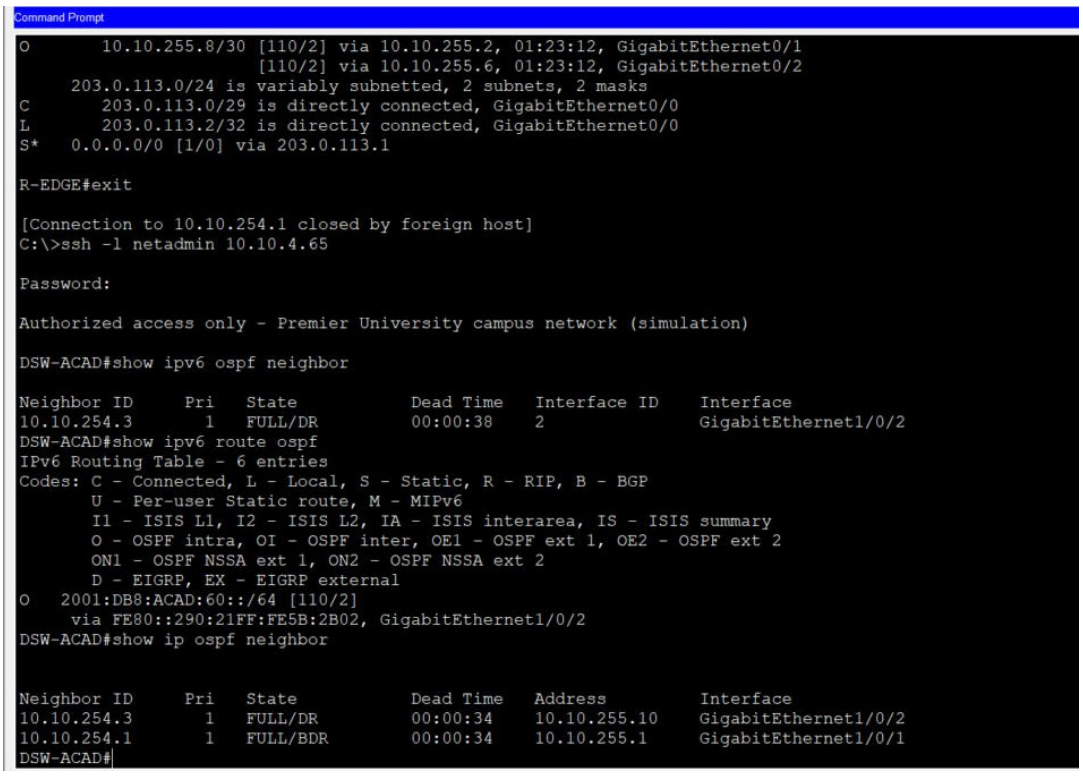


Figure 9a. DSW-ACAD: OSPFv3 neighbour, IPv6 OSPF route and OSPFv2 neighbours.

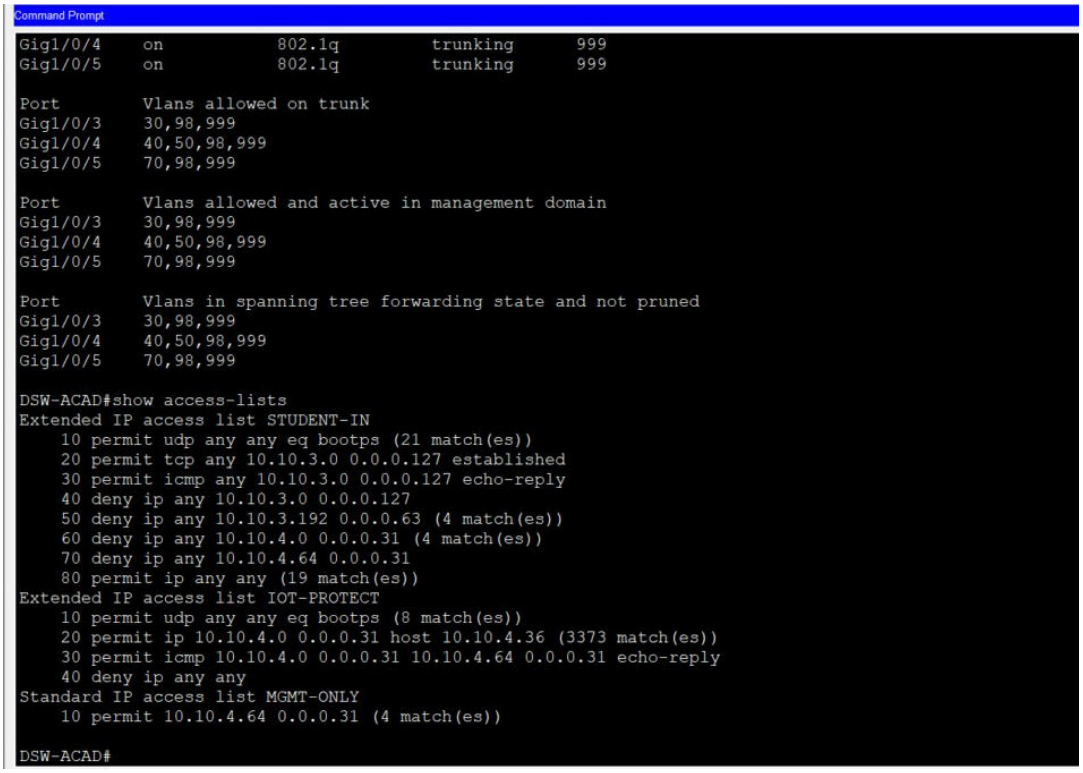


Figure 9b. DSW-ACAD: STUDENT-IN and IOT-PROTECT hit counters.

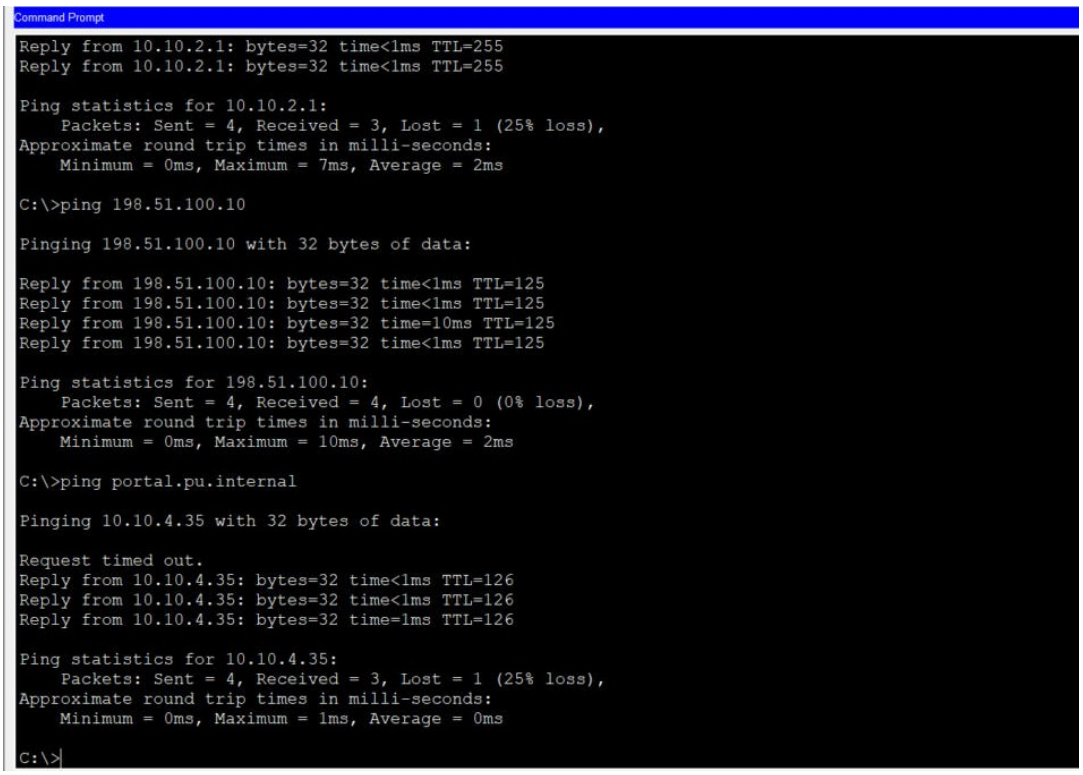


Figure 10a. PC-LAB1: gateway, Internet (through PAT) and portal-by-name reachable.

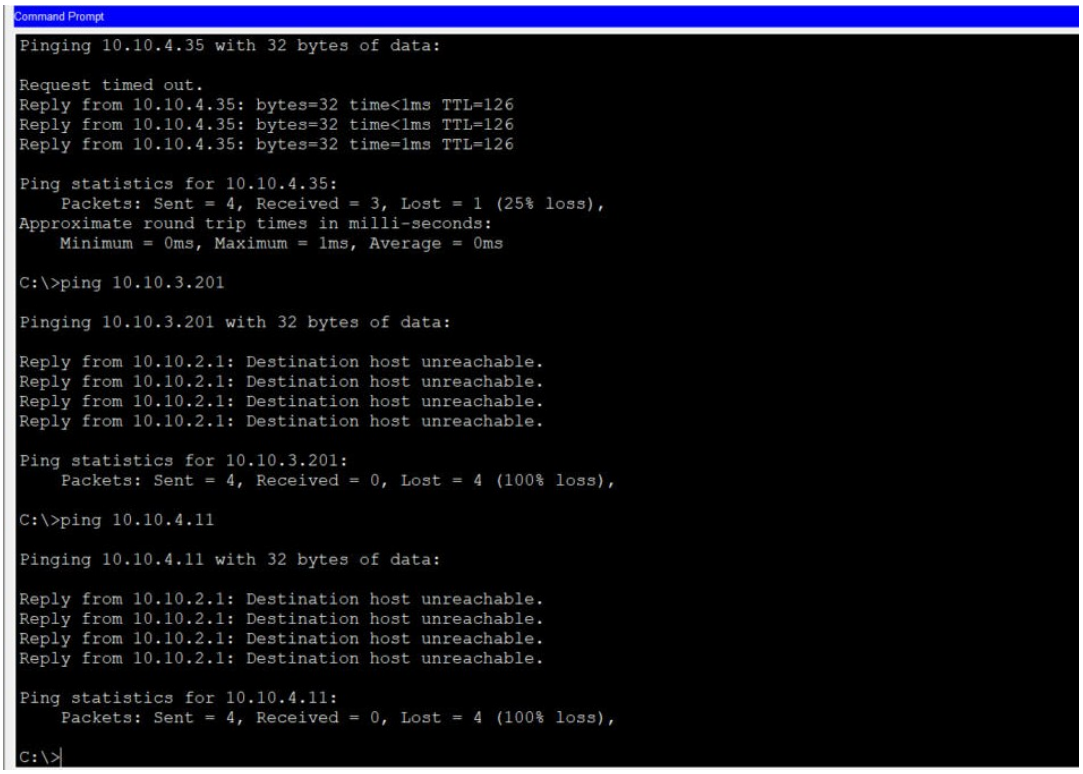


Figure 10b. PC-LAB1: registrar and IoT subnets blocked by STUDENT-IN.

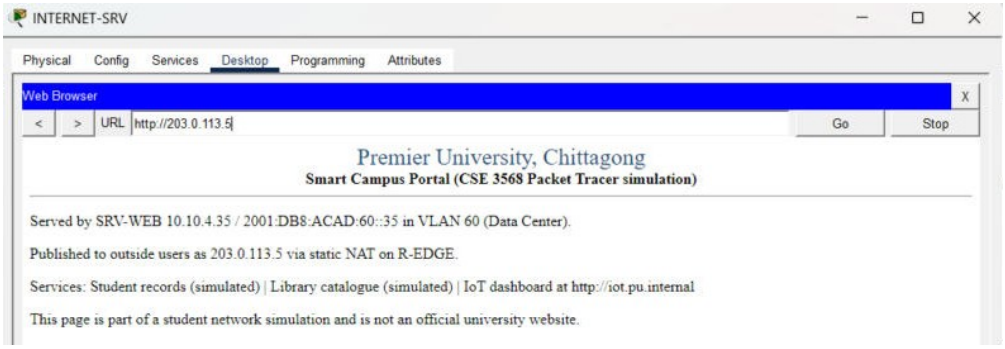


Figure 11a. INTERNET-SRV opens the portal through static NAT (203.0.113.5).

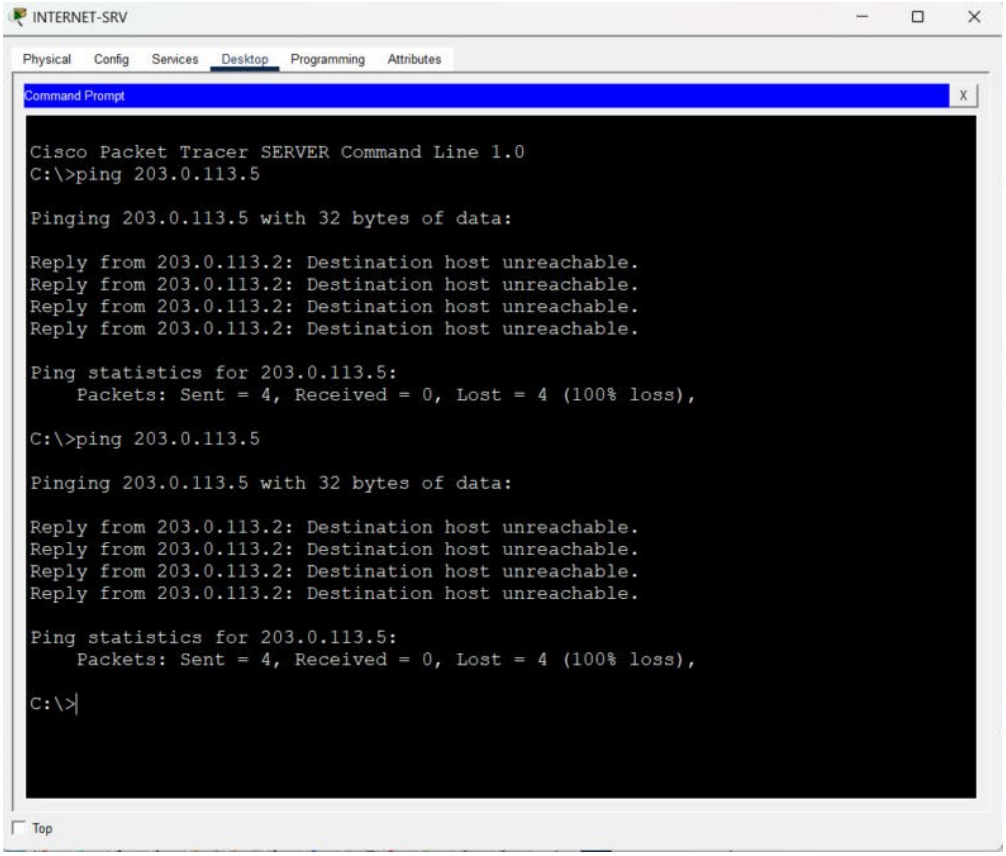


Figure 11b. INTERNET-SRV ping to 203.0.113.5 dropped by EDGE-IN.

17. Wireless Network

AP-STUDENT (AccessPoint-PT) is connected to ASW-LIB Fa0/3 in VLAN 40 and broadcasts SSID PU-STUDENT with WPA2-PSK and AES. A tablet and a smartphone were configured with the same SSID and passphrase; both associated and leased addresses from the /23 student pool. STUDENT-IN applies to them exactly as to lab PCs. A tablet was used instead of the proposed laptop because it has a built-in wireless adapter.

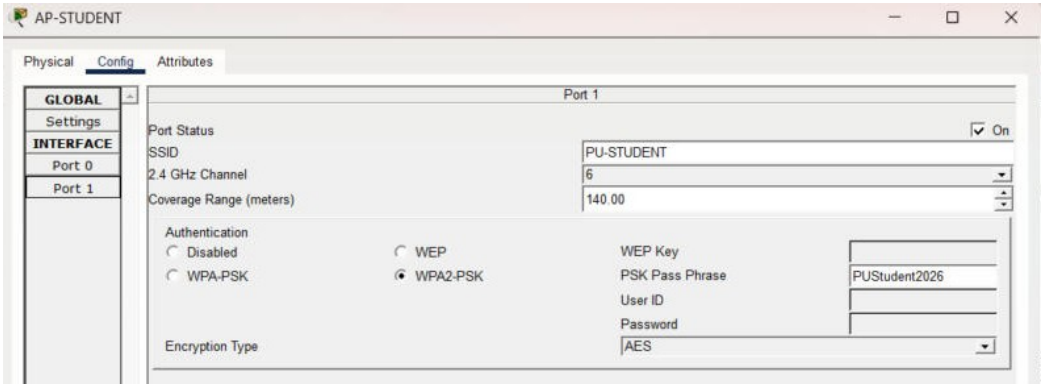


Figure 12a. AP-STUDENT security settings.

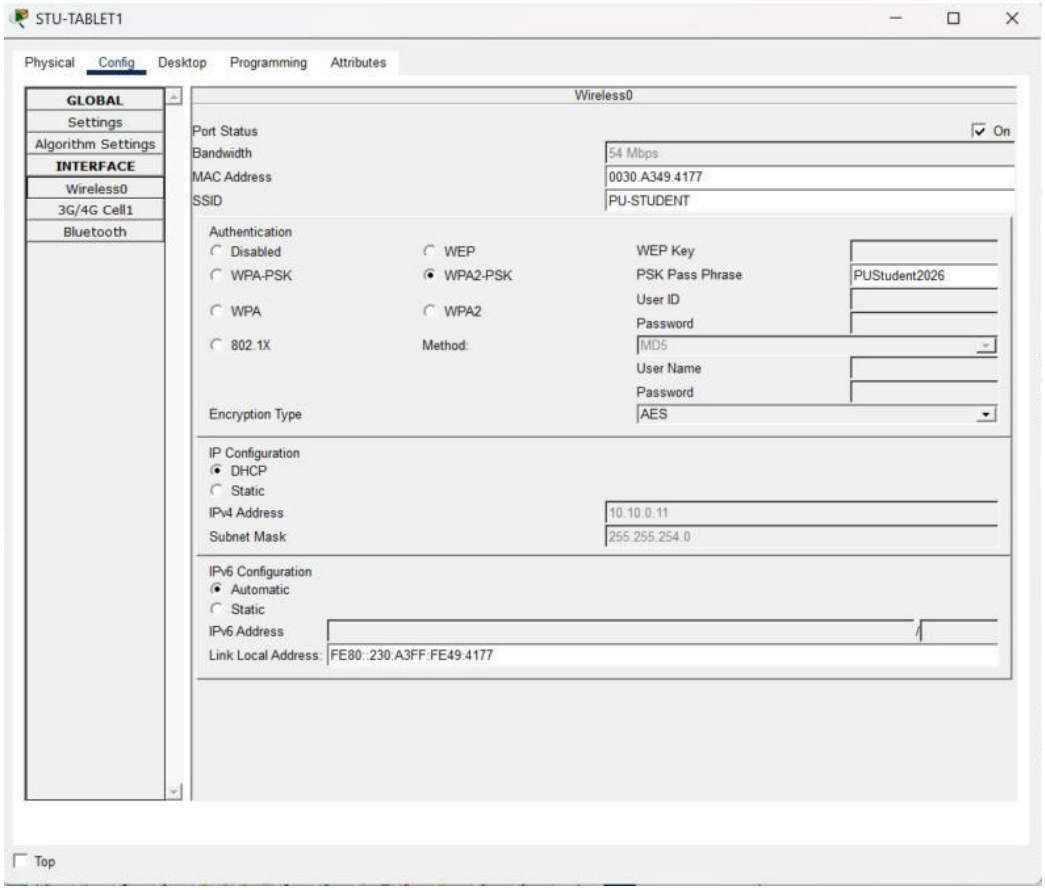


Figure 12b. STU-TABLET1 associated with WPA2-PSK/AES and leased an address from 10.10.0.0/23.

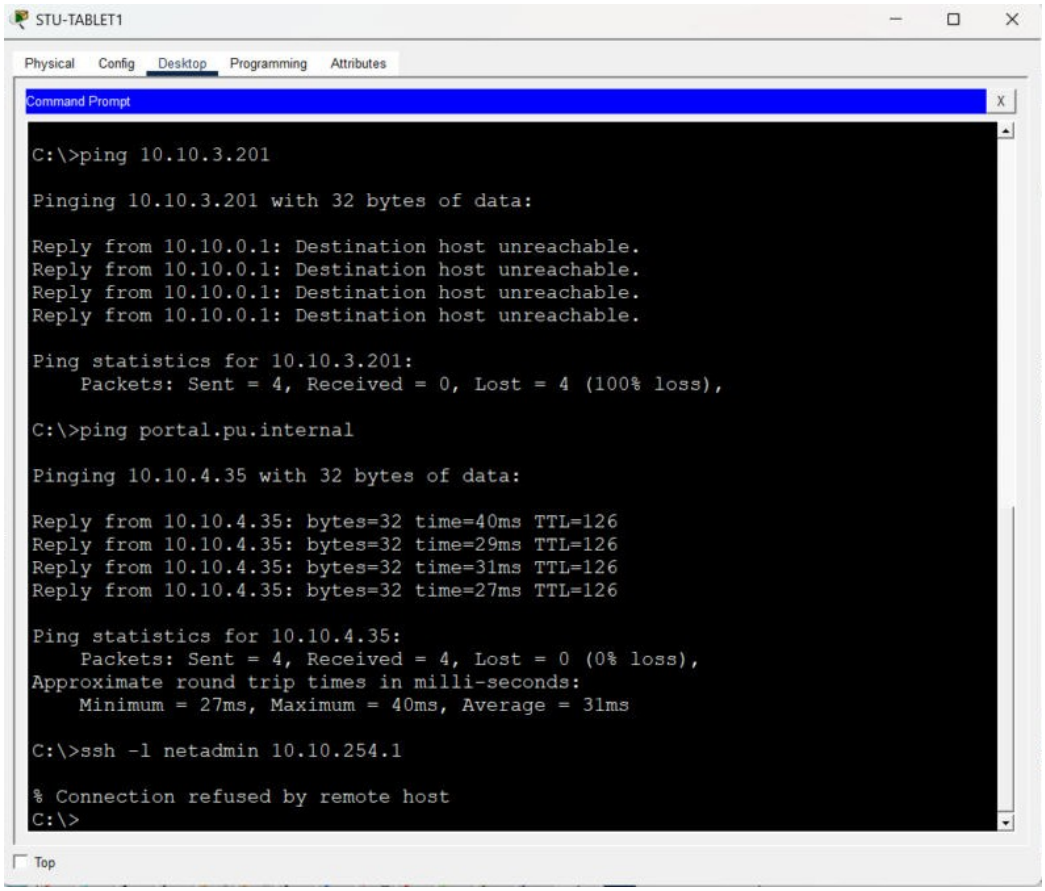


Figure 12c. STU-TABLET1: portal reachable over Wi-Fi; registrar and SSH to R-EDGE blocked.

18. IoT Automation

The four Home IoT devices were switched to the wired PT-IOT-NM-1CFE adapter, cabled to ASW-IOT (VLAN 70), addressed by DHCP and registered with SRV-IOT as remote server. An account was created on the registration server from NMS-PC, and four rules were added: R1-Lights-On / R1-Lights-Off (motion detector controls the light) and R2-Fan-High / R2-Fan-Off (temperature above 30 °C sets the fan to High; below 26 °C turns it off, the gap preventing rapid cycling). Triggering the motion detector (Alt+click) switched the light from Off to On. The temperature rule is configured but was not triggered, because the simulated ambient temperature was not changed.

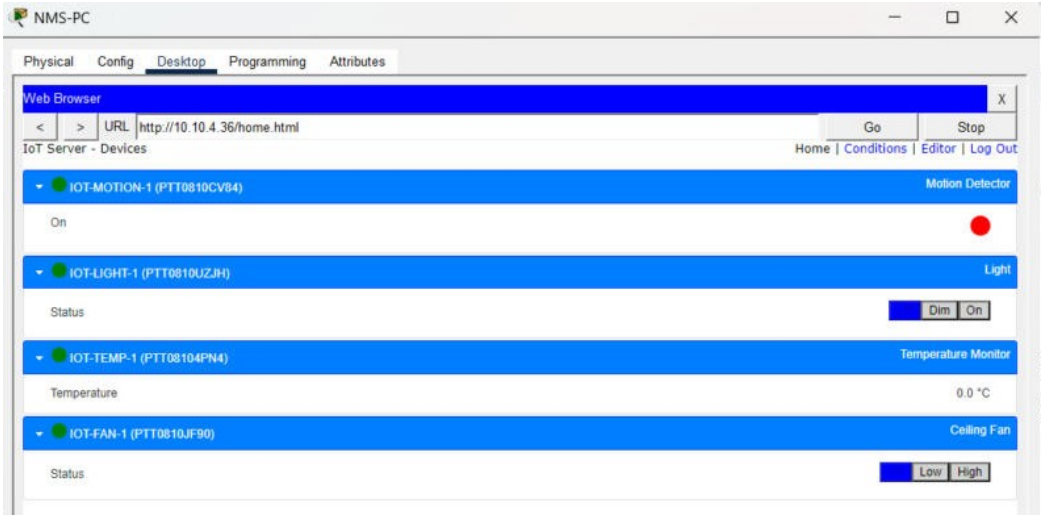


Figure 13a. All four IoT devices online on the SRV-IOT dashboard (motion detector On).

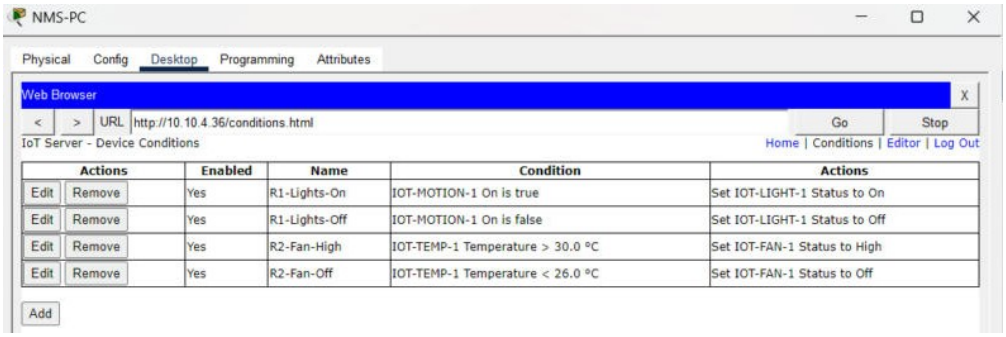


Figure 13b. Automation rules on SRV-IOT.

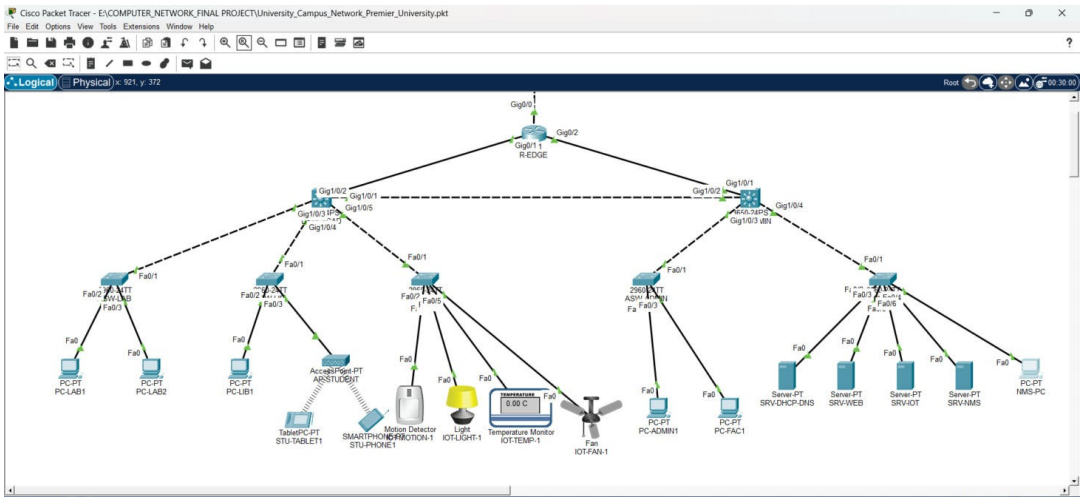


Figure 13c. After the motion trigger the smart light is on (topology view).

19. Testing Matrix

Thirty-one tests were planned against the functional requirements. Each was run in the simulation unless marked NOT TESTED. Screenshot numbers refer to files in the Screenshots folder.

ID	Area	Test	Expected	Result
T01	Routing	OSPF adjacencies — R-EDGE: show ip ospf neighbor	FULL with 10.10.254.2 and 10.10.254.3	PASS
T02	Routing	All campus subnets learned — R-EDGE: show ip route	O routes for every VLAN subnet + S* default	PASS
T03	IPv6	OSPFv3 adjacency and route — DSW-ACAD: show ipv6 ospf neighbor / show ipv6 route ospf	Neighbour FULL; 2001:DB8:ACAD:60::/64 learned	PASS
T04	IPv6	End-to-end IPv6 — PC-LIB1: ping 2001:DB8:ACAD:60::35	Replies	PASS
T05	IPv6	SLAAC on VLAN 50 — PC-LIB1 IPv6 = Automatic	Global address from RA	FAIL
T06	DHCP	Relay to VLAN 30 — PC-LAB1: ipconfig /all	10.10.2.x/24 from 10.10.4.34	PASS
T07	DHCP	Relay to VLANs 10/20/50 — PC-ADMIN1, PC-FAC1, PC-LIB1 DHCP	Address in own VLAN	PASS
T08	DHCP	Relay to IoT VLAN 70 through IOT-PROTECT — IOT-MOTION-1 DHCP	10.10.4.x/27	PASS
T09	VLAN/Trunk	VLANs and 802.1Q trunks — show vlan brief / show interfaces trunk (DSW-ACAD, DSW-ADMIN, ASW-DC)	Correct VLANs; trunks native 999, pruned lists	PASS
T10	Connectivity	Default gateway — PC-LAB1: ping 10.10.2.1	Replies	PASS
T11	DNS/Inter-VLAN	Name resolution + server reach — PC-LAB1: ping portal.pu.internal	Resolves to 10.10.4.35, replies	PASS
T12	Web	Campus portal — PC-LAB1 browser http://portal.pu.internal	Portal page	PASS
T13	ACL	Student -> Registrar blocked — PC-LAB1: ping 10.10.3.201	Fail	PASS
T14	ACL	Student -> IoT blocked — PC-LAB1: ping 10.10.4.11	Fail	PASS
T15	NAT/PAT	Inside user to Internet — PC-LAB1: ping 198.51.100.10	Replies (ISP has no route to 10.10.0.0/16)	PASS
T16	Static NAT	Outside access to portal — INTERNET-SRV browser http://203.0.113.5	Portal page	PASS
T17	ACL	Unsolicited inbound ICMP blocked — INTERNET-SRV: ping 203.0.113.5	Fail	PASS
T18	Wireless	WPA2-AES association + DHCP — STU-TABLET1 / STU-PHONE1	Associate, lease 10.10.0.x	PASS
T19	Wireless	Wi-Fi client reaches services; ACL applies — STU-TABLET1: ping portal.pu.internal, ping 10.10.3.201	Portal replies; registrar blocked	PASS
T20	Wireless	Wrong passphrase rejected — Client with wrong key	No association	NOT TESTED
T21	Management	SSH v2 from management VLAN — NMS-PC: ssh -l netadmin to R-EDGE, DSW-ACAD, DSW-ADMIN, ASW-DC	Login	PASS
T22	Management	SSH from non-management host blocked — STU-TABLET1: ssh -l netadmin 10.10.254.1	Refused	PASS
T23	Management	Console enable on R-EDGE — R-EDGE CLI tab: enable	Prompt accepts secret	FAIL
T24	IoT	Devices registered — SRV-IOT dashboard from NMS-PC	4 devices online	PASS
T25	IoT	Motion -> light rule — Alt+click IOT-MOTION-1	Light turns On	PASS
T26	IoT	Temperature -> fan rule — Raise ambient temperature above 30 C	Fan High	NOT TESTED
T27	Routing	Failover when L3 is shut — shutdown G1/0/2 on DSW-ACAD, traceroute	Traffic via R-EDGE	NOT TESTED
T28	ACL	Faculty-initiated session to lab allowed — PC-FAC1 ping PC-LAB1	Replies	NOT TESTED
T29	Switching	Port-security / BPDU Guard — Connect extra device / switch	Port err-disabled	NOT TESTED
T30	File	Native .pkt saved — File > Save / Save As	.pkt written by Packet Tracer	PASS
T31	File	Reopen .pkt and re-test — File > Open	Topology and configs persist	NOT TESTED

20. Actual Test Results

ID	Observed result	Evidence
T01	Both FULL; DSW-ACAD and DSW-ADMIN each also show 2 FULL neighbours	02, 05, 08
T02	O routes for 10.10.0.0/23 ... 10.10.4.80/28; S* 0.0.0.0/0 via 203.0.113.1	04
T03	10.10.254.3 FULL on G1/0/2; O 2001:DB8:ACAD:60::/64 [110/2]	05

ID	Observed result	Evidence
T04	4/4 replies, 0% loss	16
T05	No global address obtained; static 2001:DB8:ACAD:50::10 used instead	16
T06	10.10.2.11 /24, gw 10.10.2.1, DHCP server 10.10.4.34, DNS 10.10.4.34	12
T07	10.10.3.201 /26, 10.10.3.11 /25, 10.10.3.136 /26	10
T08	10.10.4.11 /27; IOT-PROTECT bootps entry matches	07
T09	All VLANs active; trunks on, 802.1q, native 999, allowed lists as designed	06, 08, 09
T10	3/4 replies (first lost to ARP)	13
T11	Resolved; 3/4 replies (first lost to ARP)	13
T12	Page displayed	15
T13	Destination host unreachable from 10.10.2.1; STUDENT-IN deny 10.10.3.192 matches	14, 07
T14	Destination host unreachable; STUDENT-IN deny 10.10.4.0 matches	14, 07
T15	4/4 replies; NAT-USERS permit 8 matches; 4 expired dynamic translations	13, 02, 03
T16	Page displayed; translation tcp 203.0.113.5:80 <-> 10.10.4.35:80 listed	23, 02
T17	Destination host unreachable from 203.0.113.2; EDGE-IN deny matches	24, 03
T18	Tablet 10.10.0.11 /23 (re-leased; earlier .12), phone 10.10.0.13 /23	17, 18
T19	Portal 4/4 replies; registrar blocked by 10.10.0.1	19
T20	Not run	-
T21	Logged in to all four	02, 05, 08, 09
T22	% Connection refused by remote host	19
T23	Secret rejected 3x when typed through the remote-control session; same secret works over SSH. Cause not determined.	-
T24	IOT-MOTION-1, IOT-LIGHT-1, IOT-TEMP-1, IOT-FAN-1 online	20
T25	Light changed from Off to On; dashboard shows motion On	20, 22
T26	Rule configured; ambient temperature not changed	21
T27	Not run	-
T28	Not run	-
T29	Configured only	-
T30	Binary project file saved to E:\COMPUTER_NETWORK_FINAL PROJECT	-
T31	Not run (file was re-saved after this session's checks)	-

Summary: 23 PASS, 2 FAIL, 6 NOT TESTED. The first lost ping in T10 and T11 is ARP resolution on first contact, not packet loss on an established path.

21. Screenshots and Evidence

All screenshots were captured at full screen resolution from the running Packet Tracer application (CLI font set to 12 pt for legibility) and are stored in the Screenshots folder; they are cropped to the relevant window but otherwise unedited:

- 01_Topology_Logical_View.png
- 02_R-EDGE_OSPF_Neighbors_and_NAT.png
- 03_R-EDGE_Access_Lists_Counters.png
- 04_R-EDGE_Routing_Table.png
- 05_DSW-ACAD_OSPFv2_OSPFv3_Neighbors_IPv6_Route.png
- 06_DSW-ACAD_VLANs_and_Trunks.png
- 07_DSW-ACAD_ACL_Counters.png
- 08_DSW-ADMIN_OSPF_Neighbors_and_VLANs.png
- 09_ASW-DC_VLANs_and_Trunk.png
- 10_SRV-DHCP-DNS_DHCP_Pools.png
- 11_SRV-DHCP-DNS_DNS_Records.png
- 12_PC-LAB1_ipconfig_DHCP_Lease.png
- 13_PC-LAB1_Gateway_Internet_DNS_Ping.png

- 14_PC-LAB1_ACL_Blocks_Registrar_and_IoT.png
- 15_PC-LAB1_Campus_Portal_by_Name.png
- 16_PC-LIB1_IPv6_Ping_to_SRV-WEB.png
- 17_AP-STUDENT_WPA2_AES.png
- 18_STU-TABLET1_Wireless_DHCP.png
- 19_STU-TABLET1_Portal_OK_ACL_and_SSH_Blocked.png
- 20_IoT_Dashboard_Devices_Live_State.png
- 21_IoT_Automation_Rules.png
- 22_IoT_Motion_Light_On_Topology.png
- 23_INTERNET-SRV_Static_NAT_Portal.png
- 24_INTERNET-SRV_Ping_Blocked_by_EDGE-IN.png

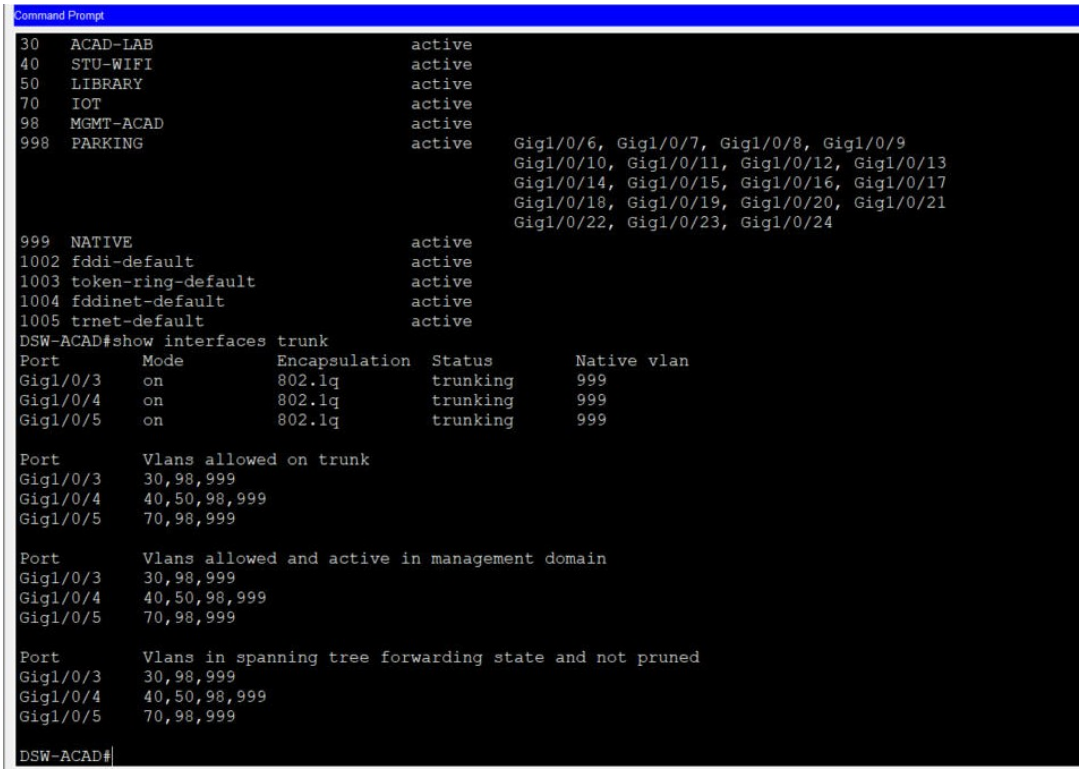


Figure 14a. DSW-ACAD: VLANs and 802.1Q trunks (native 999).

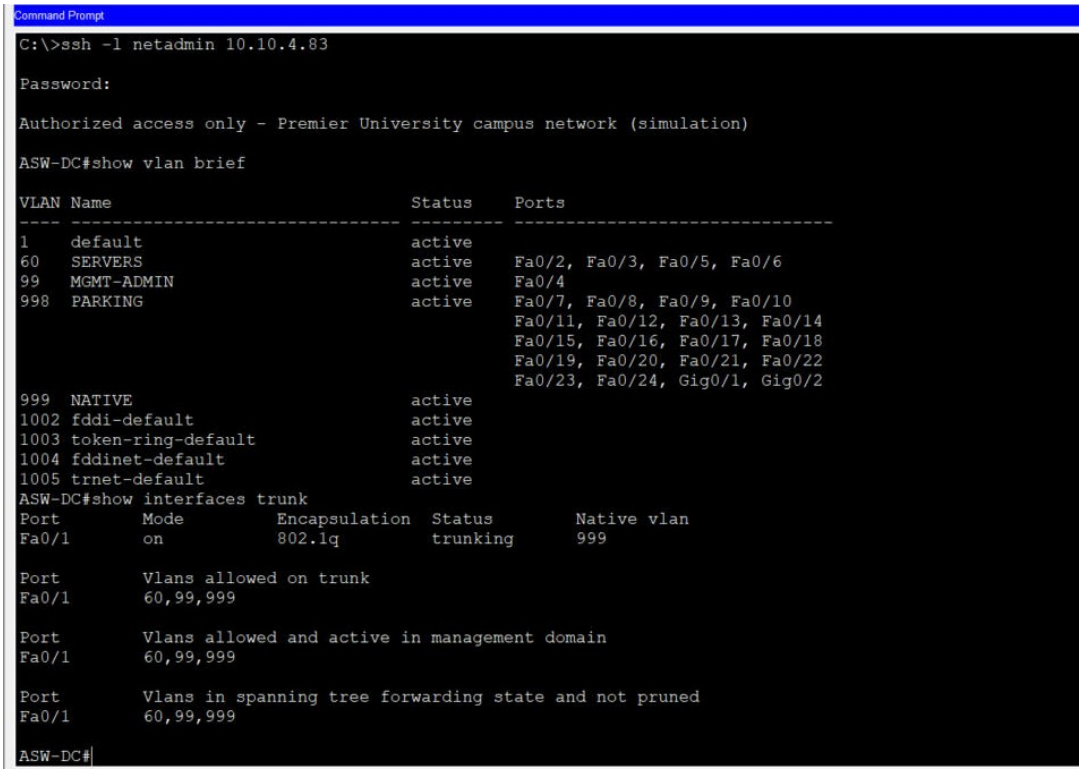


Figure 14b. ASW-DC: VLAN port membership and trunk.

22. Limitations and Deviations from the Proposal

- Edge router changed from ISR4331 to Cisco 2911: in Packet Tracer 9.0 the ISR4331's third port (G0/0/2) is SFP-only and refused the copper link to DSW-ADMIN.
- Actual ports differ from the proposal: routed links use G0/1, G0/2 on R-EDGE and G1/0/1 (uplink), G1/0/2 (L3) on the 3650s; access-switch uplinks are 2960 Fa0/1 (100 Mbps).
- PC-LIB1 did not receive an IPv6 address by SLAAC; a static IPv6 address was used (T05 FAIL). The cause (e.g. Router Advertisement behaviour of the 3650 SVI in Packet Tracer) was not investigated further.
- The R-EDGE console rejected the enable secret three times when typed through the remote-control session, although the identical secret worked over SSH (T23 FAIL). Verification on R-EDGE was therefore done over SSH from NMS-PC.
- The ICMP keyword time-exceeded is not supported by Packet Tracer 9.0, so traceroute replies from outside are not explicitly permitted by EDGE-IN.
- Not tested: OSPF failover, Wi-Fi wrong-key rejection, faculty-to-lab reachability, port-security/BPDU-guard violations, the temperature/fan rule, and reopening the saved file.
- SRV-NMS syslog/NTP is enabled but no device points at it yet; WPA2-Enterprise was not attempted.
- Device configurations in Device_Configurations are the scripts that were applied, not show running-config exports.

23. Conclusion

The project produced a working Packet Tracer model of a segmented, dual-stack smart campus network. OSPF and OSPFv3 converged across a redundant three-node routed core; DHCP relay and DNS served every client VLAN from one server; the security policy blocked students from administrative and IoT segments while keeping shared services and the Internet reachable through PAT; the portal was published safely through static NAT; WPA2 Wi-Fi clients joined the student VLAN; and an IoT rule switched classroom lighting on occupancy. The failed and untested items listed above are the natural next steps, starting with SLAAC on VLAN 50, an OSPF failover test and triggering the ventilation rule.

24. References

[1] Cisco Networking Academy, “Cisco Packet Tracer,” Cisco Systems. https://www.netacad.com/cisco-packet-tracer

[2] Cisco Systems, Cisco IOS configuration guides: OSPF, NAT, Access Control Lists, IPv6. https://www.cisco.com

[3] Y. Rekhter et al., “Address Allocation for Private Internets,” RFC 1918, IETF, 1996.

[4] J. Moy, “OSPF Version 2,” RFC 2328, IETF, 1998.

[5] R. Coltun, D. Ferguson, J. Moy and A. Lindem, “OSPF for IPv6,” RFC 5340, IETF, 2008.

[6] G. Huston, A. Lord and P. Smith, “IPv6 Address Prefix Reserved for Documentation,” RFC 3849, IETF, 2004.

[7] J. Arkko, M. Cotton and L. Vegoda, “IPv4 Address Blocks Reserved for Documentation,” RFC 5737, IETF, 2010.

[8] P. Srisuresh and K. Egevang, “Traditional IP Network Address Translator,” RFC 3022, IETF, 2001.

[9] R. Droms, “Dynamic Host Configuration Protocol,” RFC 2131, IETF, 1997.

[10] IEEE Std 802.1Q-2022; IEEE Std 802.11-2020.

25. Appendix: Device Configurations

The applied configuration scripts follow. Lab-only credentials are shown because they are part of the simulation; change them before any reuse.

ISP-R

```
! ISP-R - configuration script applied through the Packet Tracer IOS CLI (Paste) on 29 Oct 2026.
! This is the script that was entered, not a 'show running-config' export. Lab-only credentials: change
before reuse.
!
enable
configure terminal
hostname ISP-R
no ip domain-lookup
banner motd #Simulated ISP router - CSE 3568 campus network project#
interface GigabitEthernet0/0
 description WAN link to campus R-EDGE G0/0
 ip address 203.0.113.1 255.255.255.248
 no shutdown
interface GigabitEthernet0/1
 description Simulated Internet server LAN
 ip address 198.51.100.1 255.255.255.0
 no shutdown
end
write memory
```

R-EDGE

```
! R-EDGE - configuration script applied through the Packet Tracer IOS CLI (Paste) on 29 Oct 2026.
! This is the script that was entered, not a 'show running-config' export. Lab-only credentials: change
before reuse.
!
enable
configure terminal
hostname R-EDGE
no ip domain-lookup
ip domain-name pu.internal
service password-encryption
enable secret PULab2026
username netadmin privilege 15 secret PULab2026
banner motd #Authorized access only - Premier University campus network (simulation)#
interface Loopback0
 description OSPF router-ID
 ip address 10.10.254.1 255.255.255.255
interface GigabitEthernet0/0
 description WAN to ISP-R G0/0 (NAT outside)
 ip address 203.0.113.2 255.255.255.248
 ip nat outside
 no shutdown
interface GigabitEthernet0/1
 description L1 to DSW-ACAD G1/0/1
 ip address 10.10.255.1 255.255.255.252
 ip nat inside
```

```
 no shutdown
interface GigabitEthernet0/2
 description L2 to DSW-ADMIN G1/0/1
 ip address 10.10.255.5 255.255.255.252
 ip nat inside
 no shutdown
router ospf 10
 router-id 10.10.254.1
 network 10.10.255.0 0.0.0.3 area 0
 network 10.10.255.4 0.0.0.3 area 0
 network 10.10.254.1 0.0.0.0 area 0
 default-information originate
ip route 0.0.0.0 0.0.0.0 203.0.113.1
ip access-list standard NAT-USERS
 deny 10.10.4.0 0.0.0.31
 deny 10.10.4.64 0.0.0.31
 permit 10.10.0.0 0.0.255.255
ip nat inside source list NAT-USERS interface GigabitEthernet0/0 overload
ip nat inside source static 10.10.4.35 203.0.113.5
ip access-list extended EDGE-IN
 permit tcp any host 203.0.113.5 eq www
 permit tcp any any established
 permit udp any eq domain any
 permit icmp any any echo-reply
 permit icmp any any unreachable
 ! "permit icmp any any time-exceeded" was rejected by Packet Tracer 9.0 (keyword not supported) and is NOT
in the running config
 deny ip any any
interface GigabitEthernet0/0
 ip access-group EDGE-IN in
ip access-list standard MGMT-ONLY
 permit 10.10.4.64 0.0.0.31
line con 0
 logging synchronous
line vty 0 4
 access-class MGMT-ONLY in
 login local
 transport input ssh
exit
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
end
write memory
```

DSW-ACAD

```
! DSW-ACAD - configuration script applied through the Packet Tracer IOS CLI (Paste) on 29 Oct 2026.
! This is the script that was entered, not a 'show running-config' export. Lab-only credentials: change
before reuse.
!
enable
configure terminal
hostname DSW-ACAD
no ip domain-lookup
ip domain-name pu.internal
service password-encryption
enable secret PULab2026
username netadmin privilege 15 secret PULab2026
banner motd #Authorized access only - Premier University campus network (simulation)#
ip routing
ipv6 unicast-routing
ipv6 router ospf 10
 router-id 10.10.254.2
exit
vlan 30
 name ACAD-LAB
vlan 40
 name STU-WIFI
vlan 50
 name LIBRARY
vlan 70
 name IOT
vlan 98
 name MGMT-ACAD
vlan 998
 name PARKING
vlan 999
 name NATIVE
exit
interface Loopback0
 description OSPF router-ID
 ip address 10.10.254.2 255.255.255.255
interface GigabitEthernet1/0/1
 description L1 to R-EDGE G0/1
```

```
 no switchport
 ip address 10.10.255.2 255.255.255.252
 no shutdown
interface GigabitEthernet1/0/2
 description L3 inter-block link
 no switchport
 ip address 10.10.255.9 255.255.255.252
 ipv6 address 2001:DB8:ACAD:FF00::1/64
 ipv6 ospf 10 area 0
 no shutdown
interface range GigabitEthernet1/0/3 - 5
 switchport mode trunk
 switchport trunk native vlan 999
 switchport nonegotiate
 no shutdown
interface GigabitEthernet1/0/3
 description Trunk to ASW-LAB Fa0/1
 switchport trunk allowed vlan 30,98,999
interface GigabitEthernet1/0/4
 description Trunk to ASW-LIB Fa0/1
 switchport trunk allowed vlan 40,50,98,999
interface GigabitEthernet1/0/5
 description Trunk to ASW-IOT Fa0/1
 switchport trunk allowed vlan 70,98,999
interface range GigabitEthernet1/0/6 - 24
 switchport mode access
 switchport access vlan 998
 shutdown
ip access-list extended STUDENT-IN
 permit udp any any eq bootps
 permit tcp any 10.10.3.0 0.0.0.127 established
 permit icmp any 10.10.3.0 0.0.0.127 echo-reply
 deny ip any 10.10.3.0 0.0.0.127
 deny ip any 10.10.3.192 0.0.0.63
 deny ip any 10.10.4.0 0.0.0.31
 deny ip any 10.10.4.64 0.0.0.31
 permit ip any any
ip access-list extended IOT-PROTECT
 permit udp any any eq bootps
 permit ip 10.10.4.0 0.0.0.31 host 10.10.4.36
 permit icmp 10.10.4.0 0.0.0.31 10.10.4.64 0.0.0.31 echo-reply
 deny ip any any
interface Vlan30
 description Gateway ACAD-LAB
 ip address 10.10.2.1 255.255.255.0
 ip helper-address 10.10.4.34
 ip access-group STUDENT-IN in
 no shutdown
interface Vlan40
 description Gateway STU-WIFI
 ip address 10.10.0.1 255.255.254.0
 ip helper-address 10.10.4.34
 ip access-group STUDENT-IN in
 no shutdown
interface Vlan50
 description Gateway LIBRARY dual-stack
 ip address 10.10.3.129 255.255.255.192
 ip helper-address 10.10.4.34
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:50::1/64
 ipv6 ospf 10 area 0
 no shutdown
interface Vlan70
 description Gateway IOT
 ip address 10.10.4.1 255.255.255.224
 ip helper-address 10.10.4.34
 ip access-group IOT-PROTECT in
 no shutdown
interface Vlan98
 description Management ACAD block
 ip address 10.10.4.65 255.255.255.240
 no shutdown
router ospf 10
 router-id 10.10.254.2
 passive-interface default
 no passive-interface GigabitEthernet1/0/1
 no passive-interface GigabitEthernet1/0/2
 network 10.10.0.0 0.0.255.255 area 0
ip access-list standard MGMT-ONLY
 permit 10.10.4.64 0.0.0.31
line con 0
 logging synchronous
line vty 0 4
 access-class MGMT-ONLY in
 login local
```

```
 transport input ssh
exit
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
end
write memory
```

DSW-ADMIN

```
! DSW-ADMIN - configuration script applied through the Packet Tracer IOS CLI (Paste) on 29 Oct 2026.
! This is the script that was entered, not a 'show running-config' export. Lab-only credentials: change
before reuse.
!
enable
configure terminal
hostname DSW-ADMIN
no ip domain-lookup
ip domain-name pu.internal
service password-encryption
enable secret PULab2026
username netadmin privilege 15 secret PULab2026
banner motd #Authorized access only - Premier University campus network (simulation)#
ip routing
ipv6 unicast-routing
ipv6 router ospf 10
 router-id 10.10.254.3
exit
vlan 10
 name ADMIN
vlan 20
 name FACULTY
vlan 60
 name SERVERS
vlan 99
 name MGMT-ADMIN
vlan 998
 name PARKING
vlan 999
 name NATIVE
exit
interface Loopback0
 description OSPF router-ID
 ip address 10.10.254.3 255.255.255.255
interface GigabitEthernet1/0/1
 description L2 to R-EDGE G0/2
 no switchport
 ip address 10.10.255.6 255.255.255.252
 no shutdown
interface GigabitEthernet1/0/2
 description L3 inter-block link
 no switchport
 ip address 10.10.255.10 255.255.255.252
 ipv6 address 2001:DB8:ACAD:FF00::2/64
 ipv6 ospf 10 area 0
 no shutdown
interface range GigabitEthernet1/0/3 - 4
 switchport mode trunk
 switchport trunk native vlan 999
 switchport nonegotiate
 no shutdown
interface GigabitEthernet1/0/3
 description Trunk to ASW-ADMIN Fa0/1
 switchport trunk allowed vlan 10,20,99,999
interface GigabitEthernet1/0/4
 description Trunk to ASW-DC Fa0/1
 switchport trunk allowed vlan 60,99,999
interface range GigabitEthernet1/0/5 - 24
 switchport mode access
 switchport access vlan 998
 shutdown
interface Vlan10
 description Gateway ADMIN-REGISTRAR
 ip address 10.10.3.193 255.255.255.192
 ip helper-address 10.10.4.34
 no shutdown
interface Vlan20
 description Gateway FACULTY
 ip address 10.10.3.1 255.255.255.128
 ip helper-address 10.10.4.34
 no shutdown
interface Vlan60
 description Gateway SERVERS dual-stack
 ip address 10.10.4.33 255.255.255.240
 ipv6 address FE80::1 link-local
```

```
 ipv6 address 2001:DB8:ACAD:60::1/64
 ipv6 ospf 10 area 0
 no shutdown
interface Vlan99
 description Management ADMIN block
 ip address 10.10.4.81 255.255.255.240
 no shutdown
router ospf 10
 router-id 10.10.254.3
 passive-interface default
 no passive-interface GigabitEthernet1/0/1
 no passive-interface GigabitEthernet1/0/2
 network 10.10.0.0 0.0.255.255 area 0
ip access-list standard MGMT-ONLY
 permit 10.10.4.64 0.0.0.31
line con 0
 logging synchronous
line vty 0 4
 access-class MGMT-ONLY in
 login local
 transport input ssh
exit
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
end
write memory
```

ASW-LAB

```
! ASW-LAB - configuration script applied through the Packet Tracer IOS CLI (Paste) on 29 Oct 2026.
! This is the script that was entered, not a 'show running-config' export. Lab-only credentials: change
before reuse.
!
enable
configure terminal
hostname ASW-LAB
no ip domain-lookup
ip domain-name pu.internal
service password-encryption
enable secret PULab2026
username netadmin privilege 15 secret PULab2026
banner motd #Authorized access only - Premier University campus network (simulation)#
vlan 30
 name ACAD-LAB
vlan 98
 name MGMT-ACAD
vlan 998
 name PARKING
vlan 999
 name NATIVE
exit
interface FastEthernet0/1
 description Trunk to DSW-ACAD G1/0/3
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 30,98,999
 switchport nonegotiate
interface range FastEthernet0/2 - 3
 description Computer lab PCs
 switchport mode access
 switchport access vlan 30
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
interface range FastEthernet0/4 - 24 , GigabitEthernet0/1 - 2
 switchport mode access
 switchport access vlan 998
 shutdown
interface Vlan1
 shutdown
interface Vlan98
 ip address 10.10.4.66 255.255.255.240
 no shutdown
ip default-gateway 10.10.4.65
ip access-list standard MGMT-ONLY
 permit 10.10.4.64 0.0.0.31
line con 0
 logging synchronous
line vty 0 15
 access-class MGMT-ONLY in
 login local
 transport input ssh
```

```
exit
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
end
write memory
```

ASW-LIB

```
! ASW-LIB - configuration script applied through the Packet Tracer IOS CLI (Paste) on 29 Oct 2026.
! This is the script that was entered, not a 'show running-config' export. Lab-only credentials: change
before reuse.
!
enable
configure terminal
hostname ASW-LIB
no ip domain-lookup
ip domain-name pu.internal
service password-encryption
enable secret PULab2026
username netadmin privilege 15 secret PULab2026
banner motd #Authorized access only - Premier University campus network (simulation)#
vlan 40
 name STU-WIFI
vlan 50
 name LIBRARY
vlan 98
 name MGMT-ACAD
vlan 998
 name PARKING
vlan 999
 name NATIVE
exit
interface FastEthernet0/1
 description Trunk to DSW-ACAD G1/0/4
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 40,50,98,999
 switchport nonegotiate
interface FastEthernet0/2
 description Library research PC
 switchport mode access
 switchport access vlan 50
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
interface FastEthernet0/3
 description AP-STUDENT (SSID PU-STUDENT)
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
interface range FastEthernet0/4 - 24 , GigabitEthernet0/1 - 2
 switchport mode access
 switchport access vlan 998
 shutdown
interface Vlan1
 shutdown
interface Vlan98
 ip address 10.10.4.67 255.255.255.240
 no shutdown
ip default-gateway 10.10.4.65
ip access-list standard MGMT-ONLY
 permit 10.10.4.64 0.0.0.31
line con 0
 logging synchronous
line vty 0 15
 access-class MGMT-ONLY in
 login local
 transport input ssh
exit
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
end
write memory
```

ASW-IOT

```
! ASW-IOT - configuration script applied through the Packet Tracer IOS CLI (Paste) on 29 Oct 2026.
! This is the script that was entered, not a 'show running-config' export. Lab-only credentials: change
before reuse.
!
```

```
enable
configure terminal
hostname ASW-IOT
no ip domain-lookup
ip domain-name pu.internal
service password-encryption
enable secret PULab2026
username netadmin privilege 15 secret PULab2026
banner motd #Authorized access only - Premier University campus network (simulation)#
vlan 70
 name IOT
vlan 98
 name MGMT-ACAD
vlan 998
 name PARKING
vlan 999
 name NATIVE
exit
interface FastEthernet0/1
 description Trunk to DSW-ACAD G1/0/5
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 70,98,999
 switchport nonegotiate
interface range FastEthernet0/2 - 5
 description Smart campus IoT devices
 switchport mode access
 switchport access vlan 70
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
interface range FastEthernet0/6 - 24 , GigabitEthernet0/1 - 2
 switchport mode access
 switchport access vlan 998
 shutdown
interface Vlan1
 shutdown
interface Vlan98
 ip address 10.10.4.68 255.255.255.240
 no shutdown
ip default-gateway 10.10.4.65
ip access-list standard MGMT-ONLY
 permit 10.10.4.64 0.0.0.31
line con 0
 logging synchronous
line vty 0 15
 access-class MGMT-ONLY in
 login local
 transport input ssh
exit
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
end
write memory
```

ASW-ADMIN

```
! ASW-ADMIN - configuration script applied through the Packet Tracer IOS CLI (Paste) on 29 Oct 2026.
! This is the script that was entered, not a 'show running-config' export. Lab-only credentials: change
before reuse.
!
enable
configure terminal
hostname ASW-ADMIN
no ip domain-lookup
ip domain-name pu.internal
service password-encryption
enable secret PULab2026
username netadmin privilege 15 secret PULab2026
banner motd #Authorized access only - Premier University campus network (simulation)#
vlan 10
 name ADMIN
vlan 20
 name FACULTY
vlan 99
 name MGMT-ADMIN
vlan 998
 name PARKING
vlan 999
 name NATIVE
exit
interface FastEthernet0/1
```

```
 description Trunk to DSW-ADMIN G1/0/3
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,99,999
 switchport nonegotiate
interface FastEthernet0/2
 description Registrar office PC
 switchport mode access
 switchport access vlan 10
interface FastEthernet0/3
 description Faculty PC
 switchport mode access
 switchport access vlan 20
interface range FastEthernet0/2 - 3
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
interface range FastEthernet0/4 - 24 , GigabitEthernet0/1 - 2
 switchport mode access
 switchport access vlan 998
 shutdown
interface Vlan1
 shutdown
interface Vlan99
 ip address 10.10.4.82 255.255.255.240
 no shutdown
ip default-gateway 10.10.4.81
ip access-list standard MGMT-ONLY
 permit 10.10.4.64 0.0.0.31
line con 0
 logging synchronous
line vty 0 15
 access-class MGMT-ONLY in
 login local
 transport input ssh
exit
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
end
write memory
```

ASW-DC

```
! ASW-DC - configuration script applied through the Packet Tracer IOS CLI (Paste) on 29 Oct 2026.
! This is the script that was entered, not a 'show running-config' export. Lab-only credentials: change
before reuse.
!
enable
configure terminal
hostname ASW-DC
no ip domain-lookup
ip domain-name pu.internal
service password-encryption
enable secret PULab2026
username netadmin privilege 15 secret PULab2026
banner motd #Authorized access only - Premier University campus network (simulation)#
vlan 60
 name SERVERS
vlan 99
 name MGMT-ADMIN
vlan 998
 name PARKING
vlan 999
 name NATIVE
exit
interface FastEthernet0/1
 description Trunk to DSW-ADMIN G1/0/4
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 60,99,999
 switchport nonegotiate
interface range FastEthernet0/2 - 3 , FastEthernet0/5 - 6
 description Data center servers
 switchport mode access
 switchport access vlan 60
 spanning-tree portfast
 spanning-tree bpduguard enable
interface FastEthernet0/4
 description NMS-PC network management workstation
 switchport mode access
 switchport access vlan 99
 spanning-tree portfast
```

```
 spanning-tree bpduguard enable
interface range FastEthernet0/7 - 24 , GigabitEthernet0/1 - 2
 switchport mode access
 switchport access vlan 998
 shutdown
interface Vlan1
 shutdown
interface Vlan99
 ip address 10.10.4.83 255.255.255.240
 no shutdown
ip default-gateway 10.10.4.81
ip access-list standard MGMT-ONLY
 permit 10.10.4.64 0.0.0.31
line con 0
 logging synchronous
line vty 0 15
 access-class MGMT-ONLY in
 login local
 transport input ssh
exit
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
end
write memory
```

End devices and services

```
End devices and services configured through the Packet Tracer GUI (Config / Services / Desktop tabs)
================================================================================================
SRV-DHCP-DNS  10.10.4.34/28  GW 10.10.4.33  DNS 10.10.4.34
  DHCP (service ON). Pools: VLAN10-ADMIN gw 10.10.3.193 start 10.10.3.201 /26 max 54
                            VLAN20-FACULTY gw 10.10.3.1 start 10.10.3.11 /25 max 116
                            VLAN30-ACAD-LAB gw 10.10.2.1 start 10.10.2.11 /24 max 244
                            VLAN40-STU-WIFI gw 10.10.0.1 start 10.10.0.11 /23 max 500
                            VLAN50-LIBRARY gw 10.10.3.129 start 10.10.3.136 /26 max 55
                            VLAN70-IOT gw 10.10.4.1 start 10.10.4.11 /27 max 20
                            (default serverPool left in place; unused because no DHCP clients exist in VLAN
60)
  DNS (service ON): portal.pu.internal A 10.10.4.35 | iot.pu.internal A 10.10.4.36 | nms.pu.internal A
10.10.4.37
                    www.internet.sim A 198.51.100.10 | portal6.pu.internal AAAA 2001:DB8:ACAD:60::35
SRV-WEB       10.10.4.35/28  GW 10.10.4.33 | 2001:DB8:ACAD:60::35/64 GW FE80::1 | HTTP index.html = campus
portal page
SRV-IOT       10.10.4.36/28  GW 10.10.4.33 | IoT registration service ON | account iotadmin / PUIoT2026
(lab only)
              Rules: R1-Lights-On  IF IOT-MOTION-1 On = true  THEN IOT-LIGHT-1 Status = On
                     R1-Lights-Off IF IOT-MOTION-1 On = false THEN IOT-LIGHT-1 Status = Off
                     R2-Fan-High   IF IOT-TEMP-1 Temperature > 30 C THEN IOT-FAN-1 Status = High
                     R2-Fan-Off    IF IOT-TEMP-1 Temperature < 26 C THEN IOT-FAN-1 Status = Off
SRV-NMS       10.10.4.37/28  GW 10.10.4.33 | Syslog service ON (no device has been pointed at it yet)
INTERNET-SRV  198.51.100.10/24 GW 198.51.100.1 | HTTP test page
NMS-PC        10.10.4.90/28  GW 10.10.4.81  DNS 10.10.4.34 (static)
PC-LAB1, PC-LAB2, PC-ADMIN1, PC-FAC1: DHCP (leases observed 10.10.2.11, 10.10.2.12, 10.10.3.201,
10.10.3.11)
PC-LIB1       DHCP IPv4 (10.10.3.136) + static IPv6 2001:DB8:ACAD:50::10/64, gateway FE80::1
AP-STUDENT    Port 1: SSID PU-STUDENT, WPA2-PSK, AES, passphrase PUStudent2026 (lab only)
STU-TABLET1, STU-PHONE1: SSID PU-STUDENT WPA2-PSK AES, DHCP (leases 10.10.0.12, 10.10.0.13)
IoT devices (IOT-MOTION-1, IOT-LIGHT-1, IOT-TEMP-1, IOT-FAN-1): adapter changed to PT-IOT-NM-1CFE (wired),
              DHCP, IoT Server = Remote Server 10.10.4.36 (iotadmin). IOT-MOTION-1 leased 10.10.4.11.
```